Overeducation Persistence Over the Lifecycle: Disentangling Frictions, Specialization and Aging

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November 2026

Abstract

The monthly persistence of overeducation among U.S. college graduates rises steadily with age, and among graduates who are overeducated at age 30, most reach a job that requires a degree within a few years, but one in six has not reached one by age 55. I study these facts with a lifecycle directed-search model in which graduates differ by field of study and accumulate field-specific experience that is only partly transferable between non-college and college jobs, firms and workers learn about match quality over time, and the work horizon shortens with age. Calibrated to the NLSY79, the model reproduces the untargeted rise of persistence with age and the speed of the early exits from overeducation. Search and learning frictions account for most of the time that graduates spend overeducated, while the partial loss of experience and the shrinking work horizon account for the rise of persistence with age.

Keywords: overeducation, educational mismatch, persistence, directed search, lifecycle, specific human capital, learning.

JEL codes: E24, I26, J24, J62, J64.

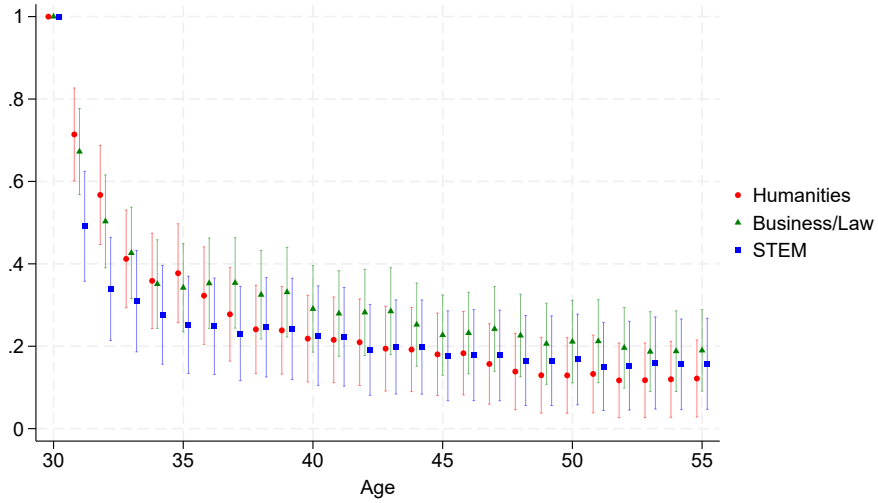
*University of Bologna. Email: aleandro.palazzo2@unibo.it. I am extremely grateful to my advisor, Giulio Zanella, for invaluable guidance and support. I thank Derek Neal, Matthew Wakefield, and Vincenzo Scrutinio for helpful comments and discussions, as well as seminar participants at the University of Chicago, University of Bologna, and Barcelona School of Economics. All errors are my own.

1 Introduction

Why do some college graduates remain in jobs that do not require their degree for many years, even decades? Between 20% and 35% of workers hold more education than the typical requirement of their job (Leuven and Oosterbeek, 2011; McGuinness et al., 2018), a condition known as overeducation. A common view treats it as a transitory mismatch: graduates accept a non-college job while searching, and move to a job that requires their degree as they gain experience. This paper documents that, for a sizable minority of graduates, overeducation is instead a long-lasting condition, and quantifies the forces that make it persistent over the lifecycle.

Figure 1 follows the U.S. college graduates of the 1979 National Longitudinal Survey of Youth (NLSY79) who are overeducated at age 30. Exits are fast at the beginning: 36% of them reach a job that requires a college degree within a year, and 67% within five years. Exits then slow down markedly: 20% of the cohort has not yet held a matched job at age 45 and 16% at age 55. Consistent with this, the monthly probability that an overeducated graduate is still overeducated one month later rises from 0.95 at age 30 to 0.98–0.99 after age 40. Overeducation thus becomes harder to leave as workers age, and those who are still overeducated in their forties are very likely to remain so.

Figure 1: Share of graduates overeducated at age 30 who have not yet held a matched job.



Note: Graduates of the NLSY79 sample who are overeducated (employed in an occupation that does not require a college degree, O*NET Job Zones 1–3) in their first month observed at age 30. The figure reports, at each age, the share of the cohort that has not held a college occupation since age 30, by field of study, with 95% confidence intervals. Weighted by the NLSY79 cross-sectional sampling weights. See Section 2 for the sample and the definitions.

To study the mechanisms behind these patterns, I develop a lifecycle model of

directed search in the spirit of Menzio et al. (2016) and Menzio and Shi (2011). Workers differ in their field of study, in their experience in the field of their job, in the quality of their current match, and in age. Directed search is a natural framework for the question: overeducated workers choose whether, and how intensely, to search for a job that requires their degree, trading off the probability of finding it against its value, so that remaining overeducated is a forward-looking choice rather than something that happens to workers. Four channels make that choice persistent. Search frictions make college jobs hard to find. Field-specific experience is only partly transferable between non-college and college jobs, so moving to a college job becomes costlier as experience in non-college jobs accumulates. Firms and workers learn the quality of a match over time, so a worker whose non-college match has turned out to be good gives up a known good match for an uncertain one when she moves. Finally, workers age: the remaining work horizon over which a better job pays off shortens.

I calibrate the model to the NLSY79, with occupational education requirements from O*NET, in two steps. The production function is identified from the age profiles of wages by field and mismatch status. The search parameters are identified from the monthly flows of graduates and non-graduates by age, from the job-tenure profiles of separations, which discipline learning about match quality, and from the change in wages at exits from overeducation, which disciplines the transferability of experience. Graduates who leave overeducation for a matched job do not suffer wage losses, and on average gain, despite having accumulated, at the median, about seven years of experience in non-college jobs; in the model, this pins down how much of that experience remains productive in a college job. Only the average exit rates from overeducation by age band are targeted: the monthly profile of persistence by age and the survival curve of Figure 1 are not. The calibrated model reproduces the rise of persistence with age and the speed of the early exits. However, it understates the long tail of the survival curve.

I then use the model to decompose persistence into its channels. I reduce each channel by the same proportion, halving the expected duration of search, the expected time to learn the quality of a match, the share of experience lost when moving between job types, and the gap between the remaining work horizon and that of a 30-year-old. Exact Shapley values attribute the change in persistence to each channel. The level of persistence and its rise with age have different sources. For STEM, business and law graduates overeducated at 30, search frictions account for 59% of the time spent in overeducation between ages 30 and 55 and learning frictions for 33%, and a 10% reduction of either lowers that time by about a year. The monthly persistence of overeducation, instead, is 1.8 percentage points higher at ages 50–55 than in the thirties, and the loss of experience (1.5 points) and the shrinking horizon (1.1 points)

account for this increase, while search frictions flatten it (-0.5 points) because they weigh more on younger workers, who have more to gain from a matched job. The loss of experience thus does not lengthen the early spells of overeducation, which end before much non-college experience accumulates, but it is the main reason why graduates who are still overeducated later in the career remain so.

For humanities graduates the decomposition is different: learning and search frictions account for almost all the time spent in overeducation (53% and 46%), and the loss of experience plays no role, since humanities graduates keep most of their experience when they move to a matched job. A new humanities match is less likely to turn out good, so overeducated humanities graduates search for a matched job mainly while the quality of their current match is still unknown, and the speed of learning governs how long they remain in that state. The rise of their persistence with age comes from the shrinking horizon and from learning.

This paper contributes to two strands of the literature. First, it adds to the work on overeducation, dating back to Freeman (1975). Much of the empirical work that followed, inspired by Duncan and Hoffman (1981), studies the effects of overeducation on wages and employment (e.g., Verdugo and Verdugo, 1989), and addresses the selection of workers into overeducation with instrumental variables (e.g., Korpi and Tåhlin, 2009) or panel methods (e.g., Bauer, 2002); see Leuven and Oosterbeek (2011) for a review. Most of this literature emphasizes cross-sectional aspects (e.g., Alba-Ramirez, 1993; Hartog, 2000), with relatively little attention to dynamics. Notable exceptions are Rubb (2003), Meroni and Vera-Toscano (2017) and Clark et al. (2017), who document substantial persistence of overeducation. Persistence is hard to reconcile with the career mobility view (Sicherman and Galor, 1990; Sicherman, 1991), in which workers accept overqualified positions as stepping stones, or with search models in which overqualification is a temporary outcome of frictions (Dolado et al., 2009; Gautier, 2002). I document the lifecycle profile of persistence and the survival of overeducation over two decades, and provide a structural decomposition of persistence into its channels over the lifecycle.

Second, the paper contributes to the literature on directed search, initiated by Montgomery (1991) and Moen (1997). The model builds on Menzio et al. (2016), who study labor reallocation over the lifecycle with search and learning frictions. The model in this paper extends their framework in several dimensions: workers differ in educational level and field of study (for college graduates); experience is field-specific and it is only partially transferable between job types; the model allows for overeducation by allowing college graduates to work in non-college jobs. Finally, the paper also relates to the work on multidimensional skill mismatch (Guvenen et al., 2020; Lise and Postel-Vinay, 2020; Baley et al., 2022), which studies how the mismatch between skills and job requirements

evolves and affects wages over the career.

The rest of the paper is organized as follows. Section 2 describes the data, documents the facts that the model is asked to explain, and presents the evidence used to discipline it. Section 3 presents the model and the channels of persistence. Section 4 describes the calibration and the identification of the parameters, Section 5 validates the model against untargeted moments, and Section 6 decomposes persistence. Section 7 concludes.

2 Data and Stylized Facts

This section describes the data, documents two facts that the model aims to explain, and presents the evidence used to discipline the model. Overeducation is quite persistent, and its persistence rises with age (Fact 1); among graduates overeducated at age 30, most reach a matched job within a few years, but a sizable minority remains in non-college jobs for decades (Fact 2). Finally, other evidence regarding the moments used in the calibration is presented.

2.1 Data and Sample

The primary dataset is the National Longitudinal Survey of Youth 1979 (NLSY79), a panel of 12,686 U.S. residents born between 1957 and 1964, interviewed annually from 1979 to 1994 and biennially thereafter. Its long panel dimension allows me to follow the same workers from age 30 to age 55, which is essential to study the long-run dynamics of overeducation. I use the 6,111 respondents of the cross-sectional sample, excluding the supplemental and military oversamples. Following Baley et al. (2022), I drop individuals with more than two years of active military service, individuals with weak labor attachment (more than ten years out of the labor force or without any reported activity), and individuals who already worked more than 1,200 hours in 1979, whose employment history before the survey is not observed. I also drop college graduates whose major is not reported and employed months without occupation information. The final sample consists of 2,672 individuals observed between ages 30 and 55 (calendar years 1987–2020), for a total of 723,999 person-months.

Monthly panel. Employment states, jobs, occupations and wages are constructed at a monthly frequency from the NLSY79 work-history files, which record the respondent's activity in every week since January 1978, including the years without an interview after 1994. I aggregate the weekly records into calendar months as in Baley et al. (2022): a worker is employed in a month when a job accounts for the most hours in

that month, and the job with the most hours is her dominant job. Jobs are linked across interviews into permanent identifiers, so that a change of employer is a change of dominant job between consecutive employed months. The occupation and the hourly wage of a month are those of its dominant job. Because the model has no participation margin, unemployment and non-participation are pooled into a single unemployment state. Transition rates are computed between consecutive calendar months, and all statistics use the NLSY79 cross-sectional sampling weights. Appendix B describes each step and the treatment of missing months.

2.2 Fields of Study and Overeducation

Workers' fields. Workers are divided into four groups according to their highest degree and, for college graduates, its major: no college degree ($f_i = 0$), humanities ($f_i = 1$), business, social sciences and law (business/law hereafter, $f_i = 2$), and STEM ($f_i = 3$). The college threshold is a bachelor's degree. Since the NLSY79 reports in each round the major at the most recently attended college, I assign the major reported when the highest degree is obtained. Table 1 lists the majors in each group.

Jobs' education requirement and field. The education requirement of an occupation is taken from its O*NET Job Zone: occupations in Job Zones 1–3, which typically require at most vocational training or an associate's degree, are classified as non-college occupations; occupations in Job Zones 4–5, which typically require a bachelor's or a graduate degree, are classified as college occupations. Each college occupation is assigned to a field according to its O*NET job family (the major group of the Standard Occupational Classification), as reported in Table 1. O*NET occupations are mapped into the harmonized NLSY79 occupation codes of Autor and Dorn (2013) through SOC 2010 and Census 2000 codes, weighting O*NET occupations by U.S. employment when several of them map into the same code; Appendix B provides additional details.

Mismatch status. A worker is classified as overeducated (O) if she holds a college degree but is employed in a non-college occupation, and matched (M) if her education level corresponds to the requirement of her occupation. The classification is vertical: a graduate employed in a college occupation of a field different from her major is matched. Workers without a degree employed in college occupations are undereducated; the model abstracts from undereducation and from horizontal mismatch, and Section 4.1 describes how these cases are mapped into the model.

Table 1: Mapping of college majors and O*NET job families into fields.

Field	College majors (worker's field f_i)	O*NET job families of college occupations (job's field f_j)
Humanities	Area studies; communications; education; fine and applied arts; foreign languages; letters; library science; theology; interdisciplinary: general liberal arts, humanities and social sciences	Arts, design, entertainment, sports and media; educational instruction and library; community and social service; personal care and service
Business/law	Business and management; counseling; home economics; law; psychology; public affairs and services; recreation; social sciences	Business and financial operations; legal; management; office and administrative support; sales and related
STEM	Agriculture and natural resources; architecture and environmental design; biological sciences; computer and information sciences; engineering; health professions; mathematics; military sciences; physical sciences; interdisciplinary: biological and physical sciences, engineering and other disciplines	Architecture and engineering; computer and mathematical; life, physical and social science; healthcare practitioners and technical; healthcare support; protective service; transportation and material moving; military specific

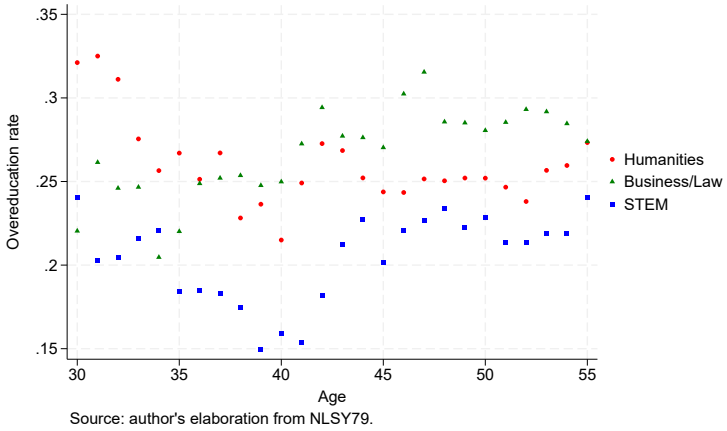
Note: College occupations are O*NET occupations in Job Zones 4–5; occupations in Job Zones 1–3 are non-college occupations ($f_j = 0$) regardless of their job family. Job families not listed (building and grounds cleaning; construction and extraction; farming, fishing and forestry; food preparation and serving; installation, maintenance and repair; production) contain no Job Zone 4–5 occupations. Several listed families consist mostly of non-college occupations and contribute few college occupations: personal care and service (1 O*NET occupation in Job Zones 4–5), office and administrative support (2), protective service (2), transportation and material moving (2), and sales and related (8); healthcare support contains none.

2.3 Stylized Facts

Overeducation is common at every stage of the career. Between 15% and 33% of employed college graduates hold a non-college job at every age between 30 and 55: 24% of graduates in their thirties and 26% of graduates aged 50–55 (Figure 2). Humanities graduates are the most affected early in the career, business/law graduates in the second half of the career, and STEM graduates are the least affected at all ages. The facts below concern how long graduates remain in this condition.

Fact 1: Overeducation is quite persistent, and its persistence rises with age. The monthly probability that an overeducated graduate is still overeducated one month later (the OO rate) is 0.949 at age 30, and averages 0.968 in the thirties, before stabilizing around 0.985 after 40 (Figure 3, bottom panel). With a constant monthly persistence of these magnitudes, a graduate overeducated in her thirties expects to remain so for about 30 months, and one overeducated after age 40 for 60–70 months. The rise in persistence comes mostly from exits to matched jobs: the monthly transition rate from overeducation to a matched job (OM) falls from 2.0% in the thirties to 0.8%

Figure 2: Share of overeducated workers by age and field.



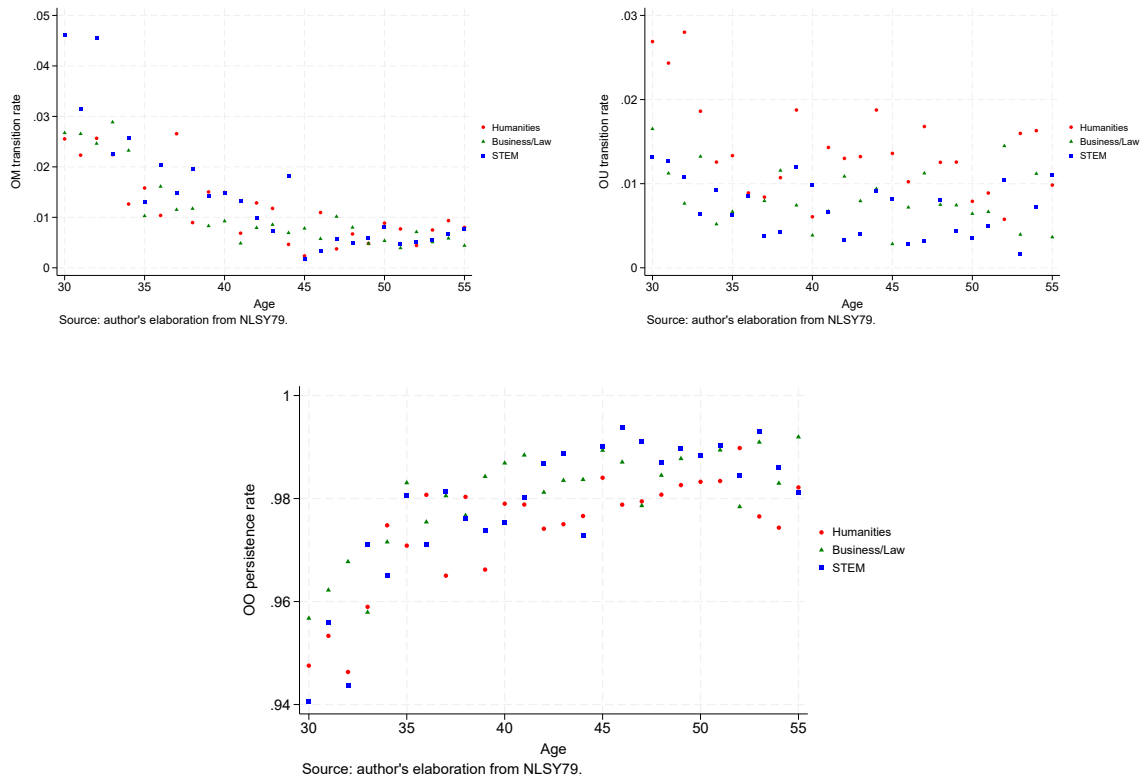
Note: Each marker is the share of employed college graduates of a given age and field of study who are overeducated, i.e. employed in a non-college occupation (O*NET Job Zones 1–3). Each graduate contributes one observation per year of age, the first month in which she is employed at that age. Fields of study are defined in Table 1. NLSY79, ages 30–55, weighted by the cross-sectional sampling weights.

in the forties and 0.6% after age 50, while the transition rate to unemployment (OU) falls less, from 1.2% to 0.8%. The decline of the OM rate is common to all fields; STEM graduates exit faster in their thirties (2.6% per month, against 1.9% for humanities and 1.8% for business/law graduates), and the differences across fields are small after age 40. This rise of persistence with age is the main object that the model aims to explain.

Fact 2: Most graduates overeducated at age 30 reach a matched job within a few years, but a sizable minority remains in non-college jobs for decades. Monthly rates do not describe how long individual workers remain overeducated. I therefore follow the graduates who are overeducated in their first month observed at age 30. Figure 1 reports the share of this cohort that has not held a matched job since age 30. Exits are fast at the beginning: 36% of the cohort reaches a matched job within one year and 67% within five years. They then slow down markedly: the share that has not yet held a matched job is 25% at age 40, 20% at age 45 and still 16% at age 55 (12% in humanities, 19% in business/law and 16% in STEM). This survival curve, with a steep initial decline followed by a long flat tail, is not targeted in the calibration and provides a demanding test of the model.

Selection on ability. A natural concern is that overeducated graduates are negatively selected on ability, so that the persistence of overeducation reflects who the overeducated are rather than the labor market forces modeled in this paper. In this section, I provide some evidence that this is not the case.

Figure 3: Monthly transition rates from overeducation to a matched job (top left), to unemployment (top right), and persistence in overeducation (bottom), by age and field.



Note: Each marker refers to the college graduates of a given age and field of study who are overeducated (employed in a non-college occupation, O*NET Job Zones 1–3) in a month, and reports the share that, in the following month, is employed in a college occupation (OM, top left), is not employed (OU, top right), or is still employed in a non-college occupation (OO, bottom); the three rates sum to one. Fields of study are defined in Table 1. NLSY79, ages 30–55, weighted by the cross-sectional sampling weights. The rates for all graduates, with 95% confidence intervals, are reported in Figure 8.

I measure ability with the scores of the Armed Services Vocational Aptitude Battery (ASVAB) and of the social scales collected in 1979–1980, when respondents were aged 14–23 and before most graduates had obtained their degree. Following Guvenen et al. (2020), I construct math, verbal, technical and social skills as the first principal component of the corresponding standardized test scores, ranked into percentiles within birth cohort; cognitive ability is defined as the average of the math and verbal percentiles, and a graduate has high ability if this average is above the median among graduates. Scores are available for 96% of the graduates in the sample.

Ability matters for who becomes overeducated and, in some fields, for how fast the overeducated leave, but not for the decline of exits with age (Table 2). Overeducated graduates score 6–9 percentiles lower than matched graduates of the same field, and low-ability graduates are overeducated more often in every field (31% of their employment

against 18% for high-ability graduates). Once overeducated, high-ability graduates move to a matched job faster in their thirties (2.5% against 1.7% per month, pooling all fields), a gap concentrated among humanities graduates (1.3 percentage points, s.e. 0.5); after age 40 the gap is smaller (0.2 points, s.e. 0.1) and significant only for business/law graduates (0.3 points, s.e. 0.15). Ability does not affect exits to unemployment (a gap of 0.05 points, s.e. 0.16, over all ages), and the OO rate of high-ability graduates is lower by only 0.4 points (s.e. 0.2). The decline of exits with age, instead, is common to both groups: the OM rate falls by about two thirds between ages 30–39 and 40–55 for low- and high-ability graduates alike, and the share of high-ability graduates among the overeducated is almost the same in the two age bands (39% and 38%). Holding this share at its value in the thirties leaves the decline of the OM rate unchanged (1.30 percentage points), so that ability composition accounts for less than 1% of the decline in every field. Measured ability also explains little of the overeducation wage penalty: once the experience profiles of college and non-college jobs are allowed to differ, controlling for the four ability percentiles changes the penalty by 1.5–3.3 log points (Table 3). These results support abstracting from ability heterogeneity in a model of the increase of persistence with age: selection on measured ability shapes entry into overeducation and, in some fields, the level of exits from it, but it does not account for the decline of exits with age, which is the focus of this paper.

Unobserved heterogeneity. Graduates may also differ in characteristics that are not measured. If some of them leave overeducation slowly for reasons that do not change over the career, the slow leavers accumulate in the stock of overeducated workers as the cohort ages, and persistence rises with age even if the exit rate of each graduate is constant. I test this explanation with the repeated spells of overeducation of the same graduates, following Heckman and Borjas (1980). Of the 1,366 spells observed from their start, 1,089 belong to the 328 graduates with two or more spells. Table 4 relates the hazard of leaving overeducation to the age at the start of the spell, first pooling all spells and then comparing only the spells of the same graduate. The comparison within graduates holds constant every characteristic of the graduate that does not change over the career, observed or not: if slow leavers drove the age profile, it would remove it. It does not. A spell that starts ten years later ends at a rate about 23% lower, both across and within graduates (hazard ratios 0.776 and 0.763). The decline is stronger for exits to a matched job (0.561 and 0.568), and it remains when every spell is followed for at most five years, so that later spells are not mechanically shorter because the panel ends (0.680 and 0.616). Within a graduate, the age at the start of a spell and the number of earlier spells move together, so this comparison does not separate the effect of age from that of earlier spells; both, however, are forms of state dependence

Table 2: Ability, entry into overeducation, and exits from overeducation.

	Business/law	STEM	Humanities	All
A. Cognitive ability percentile of employed graduates				
Matched	74.9	75.8	70.0	73.9
Overeducated	65.5	69.1	64.3	66.1
Difference	-9.4	-6.7	-5.7	-7.7
	(1.9)	(2.1)	(2.4)	(1.2)
B. Entry into overeducation (%)				
Share of high-ability graduates	52.0	54.6	40.8	49.8
Overeducation rate, low ability	34.3	25.6	30.1	30.6
Overeducation rate, high ability	20.1	15.2	17.9	18.0
Difference high − low	-14.2	-10.4	-12.2	-12.6
	(3.3)	(3.5)	(4.0)	(2.1)
Non-college share of new jobs, low ability	41.0	27.6	32.6	34.5
Non-college share of new jobs, high ability	24.3	19.6	24.3	23.0
C. Exits from overeducation to a matched job (monthly OM rate, %)				
Low ability, ages 30–39	1.56	2.41	1.54	1.74
High ability, ages 30–39	2.16	2.73	2.87	2.47
Difference high − low, ages 30–39	0.50	0.14	1.27	0.61
	(0.33)	(0.53)	(0.46)	(0.24)
Low ability, ages 40–55	0.52	0.76	0.80	0.65
High ability, ages 40–55	0.86	0.80	0.86	0.84
Difference high − low, ages 40–55	0.34	0.02	0.09	0.20
	(0.15)	(0.22)	(0.28)	(0.11)
High-ability share of the overeducated, ages 30–39	42.9	44.9	27.8	38.9
High-ability share of the overeducated, ages 40–55	40.1	41.5	31.2	38.1

Note: NLSY79 college graduates with ASVAB scores, ages 30–55, by field of study (columns). Cognitive ability is the average of the math and verbal percentiles, ranked within birth cohort and measured in 1979–1980; high-ability graduates are those above the median among college graduates (75.5), low-ability graduates the others. Panel A: mean cognitive ability of employed graduates in college occupations (matched) and in non-college occupations (overeducated); "Difference" is the coefficient of an overeducation dummy in a regression of cognitive ability on that dummy. Panel B: share of graduates with high ability; overeducation rate: share of employed months spent in a non-college occupation; non-college share of new jobs: share of non-college occupations among the jobs that start in a month, i.e. that differ from the job held in the previous month. Panel C: monthly OM rate, the share of graduates overeducated in a month who hold a college job of their field in the following month, by age band; high-ability share of the overeducated: share of high-ability graduates among the months of overeducation in the age band. In panel B, and in panel C separately for each age band, "Difference high − low" is the coefficient of a high-ability dummy in a regression of the outcome on that dummy and age dummies. In the column "All", all regressions also include field-of-study dummies. All statistics are weighted by the cross-sectional sampling weights, except the share of high-ability graduates (unweighted share of individuals). Standard errors clustered by worker in parentheses.

that time-invariant heterogeneity cannot generate. The increase of persistence with age therefore reflects how the exit rate of a given graduate changes over her career, not the selection of slow leavers into overeducation, which supports the model's focus on age-dependent channels.

Table 3: Overeducation wage penalty and ability.

	(1)	(2)	(3)	(4)
Overeducated × humanities	-0.160	-0.140	-0.132	-0.117
	(0.046)	(0.048)	(0.054)	(0.055)
Overeducated × business/law	-0.395	-0.357	-0.346	-0.313
	(0.045)	(0.045)	(0.051)	(0.051)
Overeducated × STEM	-0.455	-0.410	-0.411	-0.383
	(0.058)	(0.055)	(0.071)	(0.068)
Age and field dummies	Yes	Yes	Yes	Yes
Ability percentiles	No	Yes	No	Yes
Field-specific experience profiles	No	No	Yes	Yes
R^2	0.155	0.198	0.200	0.235
Observations (worker-months)		238,981		
Workers		989		

Note: OLS regressions on worker-months of employed NLSY79 college graduates with ASVAB scores and a valid wage, ages 30–55. The dependent variable is the log real monthly wage (hourly wage of the dominant job times 160 hours, deflated by the CPI). Each coefficient is the log wage gap between graduates of a field employed in a non-college occupation (overeducated) and graduates of the same field employed in a college occupation (matched). All columns include age and field-of-study dummies. Ability percentiles: math, verbal, technical and social percentiles, ranked within birth cohort. Field-specific experience profiles: separate quadratics in the years of experience in the field of the current job for each field of study and job type (college or non-college); experience is centered at the mean of overeducated graduates (13.4 years), so that in columns (3) and (4) the coefficients are the gaps at that experience level. Weighted by the cross-sectional sampling weights; standard errors clustered by worker in parentheses.

Table 4: Age at the start of a spell and exits from overeducation, across and within graduates.

Hazard ratio of leaving overeducation, per 10 years of age at the start of the spell		
	Pooled	Within worker
Any exit	0.776	0.763
	(0.045)	(0.063)
Exit to a matched job	0.561	0.568
	(0.043)	(0.072)
Any exit, spells followed for at most 5 years	0.680	0.616
	(0.047)	(0.064)
Spells observed from their start	1,366	
Graduates with two or more spells	328	

Note: NLSY79 college graduates, ages 30–55, months spent overeducated (employed in a non-college occupation). A spell is a sequence of consecutive overeducated months; spells observed from their start are those preceded by an observed month out of overeducation. Cox model of the hazard of leaving overeducation (any exit, or exit to a matched job) on the spells observed from their start: hazard ratio per ten years of age at the start of the spell. "Pooled" includes field-of-study dummies; "Within worker" is stratified by worker, so that it compares only the spells of the same graduate. "Spells followed for at most 5 years": every spell is censored at 60 months, and only the spells starting before age 50 are kept. Standard errors of the hazard ratios, clustered by worker, in parentheses.

2.4 Additional Evidence

This section is dedicated to presenting additional evidence that will be used to calibrate the model.

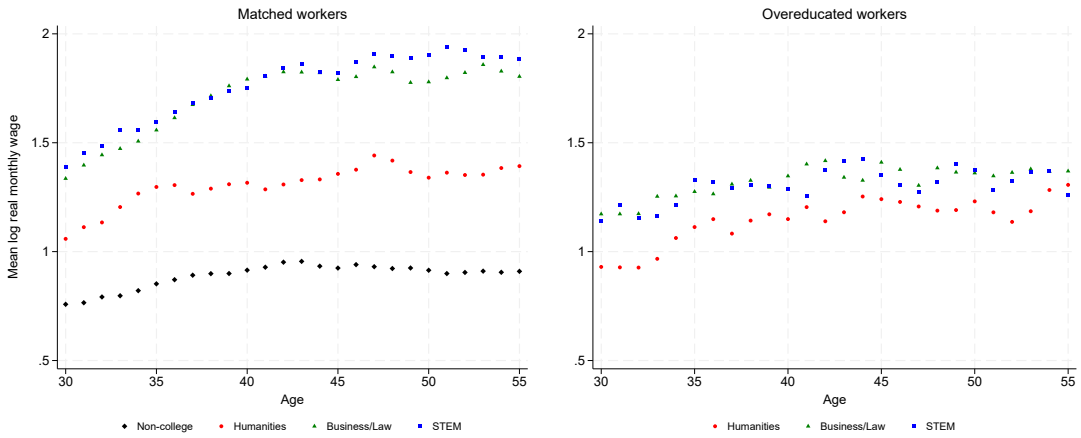
Job-tenure profiles of separations. The monthly rate at which workers leave their job falls sharply with job tenure. In college jobs, the separation rate is 5.2% in the first year of tenure for humanities jobs, 3.0% for business/law jobs and 2.7% for STEM jobs, falls to about 2% between the second and the third year and to about 1% afterwards. In non-college jobs, the rate of moves into unemployment falls from 2.5% in the first year to 0.6% after three years for non-graduates, and from 2.6% to 0.4% for overeducated graduates. As in Menzio et al. (2016), the decline of separations with tenure is informative about the speed at which firms and workers learn the quality of their match and about the probability that a new match is good.

Wage profiles by field and mismatch. Figure 4 shows the average log monthly wage of matched workers (left panel) and overeducated graduates (right panel) by field. Among matched workers, business/law and STEM graduates earn the most, with almost identical profiles that rise by about 45 log points between ages 30 and 55; humanities graduates earn 25–60 log points less, and non-college workers the least, with a profile that flattens after age 40. In every field, matched graduates earn more than their overeducated counterparts, but the overeducation penalty differs across fields: it is small and stable for humanities graduates (10–20 log points), and it widens with age for business/law graduates (from 20 to 45 log points) and especially for STEM graduates (from 25 to 60 log points). Overeducated graduates still earn 15–50 log points more than matched non-college workers, so a college degree is rewarded even in a non-college job.

Wage changes at exits from overeducation. Table 5 reports the change in the log real hourly wage at transitions from overeducation to a matched job. I measure it relative to the wage updates of graduates who remain overeducated in the same job, controlling for year effects and a quadratic in age. Moving to a matched job raises the wage by 8 log points relative to these stayers, and the gain persists one year after the transition. The gain is largest for humanities graduates (14 log points) and smaller and not statistically significant for business/law and STEM graduates (4–6 log points).

These wage changes are informative about the transferability of experience across job types. Graduates who leave overeducation have accumulated, at the median, about seven years of experience in non-college jobs. If that experience were entirely lost in a college job, the steep experience profile of wages in college occupations would

Figure 4: Age profile of wages by field and mismatch status.



Source: author's elaboration from NLSY79.

Note: Each marker is the mean log real monthly wage of employed workers of a given age and group, computed over worker-months with a valid wage. Left panel: matched workers, i.e. workers without a college degree employed in non-college occupations and college graduates employed in college occupations, by field of study. Right panel: overeducated college graduates, i.e. employed in non-college occupations (O*NET Job Zones 1–3), by field of study. Workers without a college degree employed in college occupations are excluded. The monthly wage is the hourly wage of the dominant job times 160 hours, deflated by the CPI-U (2000 dollars) and divided by the monthly federal minimum wage of 2000 ($5.15 times 160 hours), so that a value of 0 corresponds to a full-time minimum-wage job. Fields of study are defined in Table 1. NLSY79, ages 30–55, weighted by the cross-sectional sampling weights.

imply large wage losses at these transitions; the data show no losses and, on average, gains. Experience is therefore assumed to be partly transferable between non-college and college jobs, and the wage changes at transitions discipline how much of it is transferred in each field (see Section 4.4).

3 The Model

The model extends the lifecycle directed-search model of Menzio et al. (2016) to workers who differ in field of study and in field-specific experience. Moreover, it allows for possible educational mismatch between workers and jobs. Section 3.1 describes the environment, Section 3.2 the equilibrium, and Section 3.3 the channels through which the model generates persistence.

3.1 Environment

Workers, fields and jobs. Time is discrete and a period is a month. Each worker is endowed with one indivisible unit of labor, is risk-neutral, discounts the future at

Table 5: Change in the log wage at transitions from overeducation to a matched job, relative to overeducated stayers.

	At the transition		One year after	
All fields	0.076	(0.026)	0.087	(0.029)
Humanities	0.139	(0.053)	0.134	(0.063)
Business/law	0.042	(0.038)	0.062	(0.041)
STEM	0.059	(0.040)	0.075	(0.045)

Note: A transition is a month in which a college graduate employed in a non-college occupation in the previous month (overeducated) is employed in a college occupation (matched), with the same or another employer. The comparison group consists of the months in which a graduate who remains overeducated in the same job reports a change of the nominal hourly wage; since wages are recorded at interviews, these are the wage updates that stayers receive. Each cell is the coefficient of a transition dummy in an OLS regression, on transitions and comparison months, of the wage change on the dummy, year dummies and a quadratic in age, so that it measures the wage gain at the transition relative to the wage updates of stayers, as a difference in log wages. "At the transition": change in the log real hourly wage between the month before and the month of the transition. "One year after": change between the month before the transition and twelve months after it, for workers who are still in the same job and the same mismatch status. Rows by field of study estimate the regression on the graduates of each field. Ages 30–55 at the transition, NLSY79, weighted by the cross-sectional sampling weights; standard errors clustered by worker in parentheses. The field-level coefficients at the transition are the calibration targets WOM (wage change at a move from overeducation to a matched job) of Section 4.4.

rate $\beta \in (0,1)$, and lives for T periods in the labor market, indexed by her age $t \in \{0,1,\ldots,T\}$. Education is predetermined at labor market entry: each worker has a field of study $f_i \in \{0,1,\ldots,F\}$, where $f_i = 0$ denotes no college degree and $f_i > 0$ a college degree in field f_i. Each job belongs to a field $f_j \in \{0,1,\ldots,F\}$: jobs of field $f_j = 0$ do not require a college degree, jobs of field $f_j > 0$ require a degree in that field. Since the focus is on overeducation, the model abstracts from undereducation and from horizontal mismatch: the set of feasible job fields of a worker of field f_i is $\mathcal{F}(f_i) = \{0\}$ if $f_i = 0$ and $\mathcal{F}(f_i) = \{0, f_i\}$ if $f_i > 0$. A college graduate is matched if $f_j = f_i$ and overeducated if $f_j = 0$.[1] The model abstracts from differences in ability among workers of the same field; Section 2.3 shows that measured ability affects who enters overeducation and, in some fields, how fast the overeducated leave, but not the decline of exits with age.

Production. A match between a worker and a firm has quality $a \in \{a^G, a^B\}$, which is initially unknown to both parties. The output of the match is

$$y(a, f_i, f_j, e) = \begin{cases} g(f_i, f_j, e) & \text{if } a = a^G, \\ 0 & \text{if } a = a^B, \end{cases} \tag{1}$$

[1] In the data, graduates employed in a college job of another field are assigned to a job of their own field, and workers without a degree employed in a college job to a non-college job (Section 4.1).

with

$$g(f_i, f_j, e) = \left[\rho_{f_j} + \mu^{ov}_{f_i} \mathbf{1}\{f_i > 0,\, f_j = 0\}\right] h_{f_j}(e), \tag{2}$$

where e is the worker's experience in the field of the job. The baseline productivity ρ_{f_j} depends on the field of the job, $\mu^{ov}_{f_i}$ is the productivity premium of a graduate of field f_i in a non-college job relative to a non-graduate, and

$$h_{f_j}(e) = (1 - \phi_{1,f_j}) + \phi_{1,f_j}(1 + e)^{\phi_2} \tag{3}$$

captures the returns to experience in field f_j, which may differ across job fields.[2] A new match has good quality with probability $\bar{a}_{f_j}$, which depends on the field of the job. While employed, the quality of the match is revealed with probability α in each period; until then it is denoted a^0.

Amenities. A good match in a college job of field $f_j > 0$ also yields the worker a non-pecuniary flow value χ_{f_j}, which captures the amenities of working in a job that requires her degree, such as the content of the job, career concerns and social status. The flow value of a match is therefore

$$z(a, f_i, f_j, e) = y(a, f_i, f_j, e) + \chi_{f_j}\, \mathbf{1}\{a = a^G,\, f_j > 0\}. \tag{4}$$

The amenity enters the value of the job but not its output: wages are a share of output (see the paragraph on contracts below), so χ_{f_j} is a compensating differential that does not show up in wages. It can be negative, if college jobs of field f_j have disamenities such as longer hours.

Experience. A worker accumulates experience in the field of her current job: experience increases by one in each period of employment. When a worker moves to a job of the same field, she keeps her experience. When a graduate moves between a non-college job and a job of her field, in either direction, she keeps a share $\tau_{f_i} \in [0, 1]$ of it, so that her experience in the new job is $\tau_{f_i} e$. The case $\tau_{f_i} = 0$ corresponds to fully field-specific experience, and $\tau_{f_i} = 1$ to experience that is fully general across job types. When unemployed, the worker keeps the experience of her most recent job (but it does not increase).

[2] The model abstracts from the depreciation of skills with age. A depreciation that lowers the output of every job by the same proportion does not change the choice between a non-college and a college job, which is the margin that governs persistence, and the wage profiles do not separate it from the concavity of the returns to experience: in Step 1 of the calibration, adding a proportional depreciation improves the fit of the wage profiles, but it raises the estimated returns to experience at a given age from 25–44 to 54–72 log points while leaving the growth of wages over the career, which is what the data pin down, almost unchanged (Section 4.3).

Unemployment. An unemployed worker enjoys a flow value $b_t = B_{30}\left[1 + g_b\left(\text{age}_t - 30\right)\right]$, which is common to all workers and grows with age at rate g_b. The level B_{30} equals 71% of the average output of a good non-college match of non-graduates at age 30, following the replacement ratio of Hall and Milgrom (2008).

Labor market. The labor market is organized in a continuum of submarkets indexed by the type of worker, the field of the job $f'_j \in \mathcal{F}(f_i)$, and the lifetime value x offered to the worker. Firms choose in which submarkets to open vacancies, at a flow cost $k_{f'_j}$ that depends on the field of the job; workers choose where to search. In a submarket with tightness θ (the ratio of vacancies to searching workers), a worker meets a vacancy with probability $p(\theta)$, where p is continuous and concave, with $p(0) = 0$, strictly increasing and strictly concave up to some $\bar{\theta} \in (0, \infty]$ and equal to one afterwards; a vacancy meets $q(\theta) = p(\theta)/\theta$ workers on average. Unemployed workers search with probability λ_u in each period, and employed workers with probability λ_e. Thus, overeducated workers may search on the job to find a matched job.

Exogenous matched-job offers. In each period, an overeducated graduate receives with probability π, common to all fields, an exogenous offer for a matched job. The match keeps its quality, the worker keeps the share τ_{f_i} of her experience, and she accepts the offer if it is worth more than her current job. This reduced-form channel captures the transitions to matched jobs that do not result from the channels explicitly modeled, such as internal reassignments, referrals, personal reasons or other mechanisms.

Timing. Each period has five stages. (i) Exogenous matched-job offers: an overeducated graduate receives an offer of a matched job with probability π and decides whether to accept it. (ii) Separation: an employed worker becomes unemployed with probability $d \in [\delta_{f_j}, 1]$, where δ_{f_j} is the exogenous separation rate, equal to δ_0 in non-college jobs and to δ_c in college jobs, and d is specified by the employment contract. (iii) Search: workers who did not lose their job search with probability λ_u or λ_e; firms open vacancies. (iv) Matching: a worker who meets a vacancy in submarket x receives a contract worth x and accepts it or returns to her previous position; a new match has unknown quality a^0. (v) Production: employed workers produce, consume their wage w and enjoy the amenity of their job, and unemployed workers enjoy b_t; the quality of a match of unknown quality is revealed with probability α.

Contracts. As in Menzio et al. (2016), employment contracts are complete: they specify the wage, the separation probability and the submarket in which the employed worker searches, as functions of the history of the match. Contracts therefore maximize

the joint value of the match, and the wage only splits it between the worker and the firm (see Menzio and Shi, 2011, for more details). Search, separations and all other allocations depend only on the joint value of the match and on the value promised to the worker at hiring, not on how wage payments are spread over the life of the match. The model therefore does not pin down wages, and in particular the wage of a match whose quality is still unknown. I adopt the convention that the wage in every period is a constant share of the output of a good match, $y(a^G, f_i, f_j, e)$, whether or not the quality of the match is already known.[3]

3.2 Equilibrium

Let $\boldsymbol{\omega} = (a, f_i, f_j, e)$ denote the individual state of a worker, where f_j and e refer to the most recent job for an unemployed worker, and let $\hat{\boldsymbol{\omega}} = (a, f_i, f_j, e+1)$ be the state of an employed worker in the next period. A worker of state $\boldsymbol{\omega}$ who searches in field f'_j forms a new match in state

$$\nu(\boldsymbol{\omega}, f'_j) = \big(a^0, f_i, f'_j, \varepsilon(e, f_j, f'_j)\big), \qquad \varepsilon(e, f_j, f'_j) = \begin{cases} e & \text{if } f'_j = f_j, \\ \lfloor \tau_{f_i} e \rfloor & \text{if } f'_j \neq f_j. \end{cases} \tag{5}$$

Thus, the new match has unknown quality and the experience is kept if the worker moves to a job of the same field, whereas it is partially lost otherwise.

The value of unemployment of a worker of age t is

$$U_t(\boldsymbol{\omega}) = b_t + \beta \left[U_{t+1}(\boldsymbol{\omega}) + \lambda_u R_{t+1}(\boldsymbol{\omega}) \right], \qquad R_t(\boldsymbol{\omega}) = \max_{x,\, f'_j \in \mathcal{F}(f_i)} p\big(\theta_t(\boldsymbol{\omega}, f'_j, x)\big) \left[x - U_t(\boldsymbol{\omega}) \right]. \tag{6}$$

In the current period, the worker enjoys the unemployment flow value b_t, which increases with age. In the next period, she can search for a job with probability λ_u and, if she does, she receives the expected gain R_{t+1} from searching in the optimal submarket. The expected gain is the product of the probability of meeting a vacancy, $p(\theta)$, and the expected gain from accepting it, $x - U_t(\boldsymbol{\omega})$.

The joint value of a match of known quality $a \in \{a^G, a^B\}$ is

$$V_t(\boldsymbol{\omega}) = z(a, f_i, f_j, e) + \beta \big[(1-\pi)\, W_{t+1}(\hat{\boldsymbol{\omega}}) + \pi \max \{ W_{t+1}(\hat{\boldsymbol{\omega}}),\, W_{t+1}(\hat{\boldsymbol{\omega}}^\pi) \} \big], \tag{7}$$

[3] The convention affects no flow and no value in the model, only the mapping between the model and the wage data. The alternative of paying a share of the expected output, $\bar{a}_{f_j}\, y(a^G, f_i, f_j, e)$, while the quality is unknown is rejected by the data. It implies that the wages of surviving matches in college jobs rise by 44–53 log points over the first two years of tenure, as their good quality is revealed, while in the NLSY79 the wage of graduates who stay in the same college job grows by 1.9–3.1 log points per year in the first three years of tenure and by 1.0–1.7 afterwards. It also implies wage losses of 38–42 log points at exits from overeducation, against the gains of Table 5 (Appendix C).

where $\hat{\boldsymbol{\omega}}^{\pi} = (a, f_i, f_i, \lfloor \tau_{f_i}(e+1) \rfloor)$ is the state in the matched job offered to an overeducated worker, and the flow value of employment, z, is given by equation (4). The continuation value W_{t+1} is

$$W_{t+1}(\boldsymbol{\omega}) = \max_{d \in [\delta_{f_j}, 1]} \left\{ d\, U_{t+1}(\boldsymbol{\omega}) + (1-d) \left[V_{t+1}(\boldsymbol{\omega}) + \lambda_e S_{t+1}(\boldsymbol{\omega}) \right] \right\}, \tag{8}$$

$$S_{t+1}(\boldsymbol{\omega}) = \max_{x,\, f_j' \in \mathcal{F}(f_i)} p\big(\theta_{t+1}(\boldsymbol{\omega}, f_j', x)\big) \left[x - V_{t+1}(\boldsymbol{\omega}) \right]. \tag{9}$$

The flow value of a match is given by z; in the next period, overeducated workers may receive an exogenous offer of a matched job with probability π ($\pi = 0$ for all non-overeducated workers), and they accept it if it is worth more than their current job. If they accept the offer, they move to a matched job of their field and keep a share τ_{f_i} of their experience. If they do not receive an exogenous offer (probability $1-\pi$), they move to unemployment with probability d or remain employed with probability $1-d$. If they remain employed, they may search for a new job with probability λ_e and, if they do, they receive the expected gain S_{t+1} from searching in the optimal submarket. Through this channel, the model allows overeducated workers to search on the job for a matched job. The joint value of a match of unknown quality is

$$\begin{aligned} V_t(\boldsymbol{\omega}(a^0)) = {} & \alpha \left[\bar{a}_{f_j} V_t(\boldsymbol{\omega}(a^G)) + (1 - \bar{a}_{f_j}) V_t(\boldsymbol{\omega}(a^B)) \right] + (1-\alpha)\, \bar{a}_{f_j}\, z(a^G, f_i, f_j, e) \\ & + (1-\alpha)\, \beta \big[(1-\pi)\, W_{t+1}(\hat{\boldsymbol{\omega}}(a^0)) + \pi \max \left\{ W_{t+1}(\hat{\boldsymbol{\omega}}(a^0)),\, W_{t+1}(\hat{\boldsymbol{\omega}}^{\pi}(a^0)) \right\} \big]. \end{aligned} \tag{10}$$

In the current period, the quality of the match is revealed with probability α: it is good with probability $\bar{a}_{f_j}$ and bad otherwise, and the match takes the value of a match of known quality. With probability $1-\alpha$, the quality remains unknown: the match yields the expected flow value $\bar{a}_{f_j} z(a^G, f_i, f_j, e)$, since a bad match produces nothing and gives no amenity, and in the next period the worker faces the same choices as in a match of known quality (the exogenous offer, the separation and the search on the job) with a match whose quality is still unknown.

Free entry of vacancies implies that, in every submarket,

$$k_{f_j'} \geq q\big(\theta_t(\boldsymbol{\omega}, f_j', x)\big) \left[V_t\big(\nu(\boldsymbol{\omega}, f_j')\big) - x \right], \tag{11}$$

with equality if $\theta_t(\boldsymbol{\omega}, f_j', x) > 0$.

As in Menzio et al. (2016), search is directed, so the value and policy functions do not depend on the distribution of workers across states, and the equilibrium is block recursive. Appendix A shows that a block-recursive equilibrium exists, that every recursive equilibrium is block recursive, and that value functions and market tightness are unique.

3.3 Channels of Persistence

The monthly probability that an overeducated graduate of state $\boldsymbol{\omega}$ moves to a matched job is

$$\mathrm{OM}_t(\boldsymbol{\omega}) = \pi A_t(\boldsymbol{\omega}) + [1 - \pi A_t(\boldsymbol{\omega})]\,[1 - d_t(\boldsymbol{\omega})]\,\lambda_e\, p\big(\theta_t^e(\boldsymbol{\omega})\big)\,\mathbf{1}\{f_t^e(\boldsymbol{\omega}) = f_i\}, \tag{12}$$

where $A_t \in \{0,1\}$ indicates that she accepts an offer, and f_t^e and θ_t^e are the field and the tightness of the submarket in which she searches. With the functional form used in the quantitative analysis, $p(\theta) = \min\{\theta^{1/2}, 1\}$, the probability of finding a matched job when searching for one is

$$p\big(\theta_t^e(\boldsymbol{\omega})\big) = \min\left\{\frac{D_t(\boldsymbol{\omega})}{2k_{f_i}},\, 1\right\}, \qquad D_t(\boldsymbol{\omega}) = V_t\big(a^0, f_i, f_i, \lfloor \tau_{f_i} e \rfloor\big) - V_t(\boldsymbol{\omega}), \tag{13}$$

when $D_t > 0$, and zero otherwise. The following are the channels through which the model generates persistence in overeducation.

Search frictions. The vacancy cost k_{f_i} scales the job-finding probability: halving it doubles the probability of finding a matched job, that is, halves the expected duration of search.

Loss of experience. A worker who moves to a matched job keeps only $\tau_{f_i} e$ of her experience. When $\tau_{f_i} < 1$, the experience lost grows with the experience accumulated in non-college jobs, and the surplus D_t falls as the worker ages in overeducation. This channel makes persistence rise with experience, and therefore with age: the specialization trap.

Learning frictions. A worker whose non-college match is known to be good compares it with a new match that is good only with probability $\bar{a}_{f_i}$. For low values of $\bar{a}_{f_i}$, D_t is small or negative for these workers, who stop searching. Workers in matches of unknown quality, instead, value the option to learn the quality of their current match before leaving it. Faster learning separates bad matches earlier and shortens the time spent in matches of unknown quality.

Aging. The gain from moving to a better job accrues over the remaining work horizon, so D_t falls as the worker approaches retirement (the horizon effect).

Section 6 quantifies the contribution of each channel to the persistence of overeducation.

4 Calibration and Identification

4.1 Mapping the Model to the Data

The data contain undereducated workers and college graduates employed outside their field, while the model abstracts from both. Before computing the calibration moments, the data are mapped into the model's job fields: a college graduate employed in a college occupation of any field is assigned to a job of her own field ($f_j = f_i$), and a worker without a college degree employed in a college occupation is treated as holding a non-college job ($f_j = 0$). Cross-field employment is not rare: about one third of the employment of graduates in college occupations is in an occupation of another field, and 26% of the employment of workers without a degree is in college occupations. Since the focus of the paper is on overeducation, this simplification keeps the model tractable without affecting the transitions between non-college and college jobs, which are the object of the paper. Experience is measured in the model job field: for a graduate in a college job it pools all her months in college occupations, for a graduate in a non-college job it counts her months in non-college occupations, and for a worker without a degree all her months of employment. Fields are humanities ($f = 1$), business/law ($f = 2$) and STEM ($f = 3$).

4.2 Assigned Parameters and Functional Forms

The model period is one month and workers enter the model at age 18 and retire at 65. The discount factor is set so that the annual interest rate is 4.04%, the average Federal Funds Rate between 1982 and 2020 ($\beta = 0.9967$). The search probability of unemployed workers is normalized to $\lambda_u = 1$, and the flow value of unemployment at age 30 is set to 71% of the output of a good non-college match of non-graduates at that age (Hall and Milgrom, 2008). Following Menzio et al. (2016), the matching function is $p(\theta) = \min\{\theta^{1/2}, 1\}$.[4] The kink at $\bar{\theta} = 1$ is never reached: in the calibrated model, optimal tightness stays below 0.11 (a job-finding probability of 0.32) at ages 30–55. Following the convention of Section 3, wages are a constant share of the output

[4]Under this specification $q(\theta) = \theta^{-1/2} > 1$ for $\theta < 1$, so in active submarkets a vacancy meets more than one worker per period on average. This is an implication of the normalization of vacancy units and has no bearing on the results. Consider the more general form $p(\theta) = \min\{A\theta^{1/2}, 1\}$ with $A > 0$. Substituting $\phi = A^2\theta$, the search problem $\max_{\theta\geq 0}\{-k_f\theta + p(\theta)D\}$ becomes $\max_{\phi\geq 0}\{-(k_f/A^2)\phi + \min\{\phi^{1/2}, 1\}D\}$, so only the ratio k_f/A^2 is identified from worker flows. Rescaling $(1, k_f)$ to (A, A^2k_f) for all f leaves job-finding probabilities, value and policy functions unchanged, as well as offered values $x = V - k_f/q(\theta)$, since $k_f/q(\theta) = k_f\theta/p(\theta)$ is invariant. Under the rescaled specification $q(\theta) = A^2/p(\theta) \leq 1$ in every submarket with $p(\theta) \geq A^2$, and any A below the square root of the smallest job-finding probability among active submarkets makes q a proper probability in equilibrium. As the calibration targets no vacancy data, I normalize $A = 1$ without loss of generality.

of a good match, whether or not the quality of the match is known. The remaining parameters are calibrated in two steps.

4.3 Step 1: Production Function

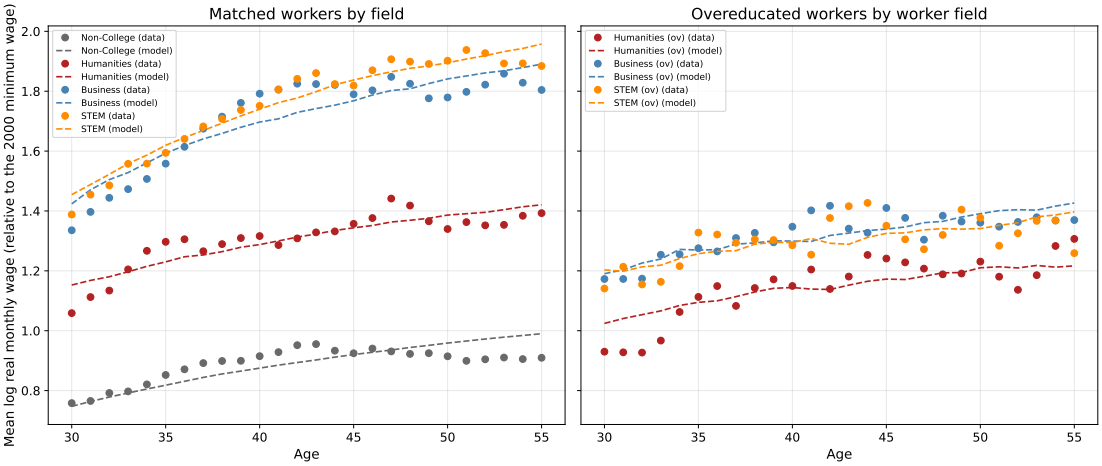
The 12 production-function parameters—the baseline productivities $\rho_0, \ldots, \rho_3$, the overeducation premia $\mu_1^{ov}, \mu_2^{ov}, \mu_3^{ov}$, the experience weights $\phi_{1,0}, \ldots, \phi_{1,3}$ and the concavity ϕ_2—are chosen to minimize the distance between the average log wages implied by the model and those of Section 2.4, for matched workers of each field and overeducated graduates of each field at each age between 30 and 55 (182 moments). For each cell and age, the model wage is computed from the output of a good match, which under this convention determines the wage of every employed worker, at the empirical distribution of field-specific experience (deciles from the NLSY79). Figure 5 shows that the model reproduces the levels and the concave age profiles of wages; Table 6 reports the parameters. Returns to experience are steeper in business/law and STEM jobs, where the output of a good match rises by 44 log points between 5 and 20 years of experience, than in non-college and humanities jobs (26 and 25 log points): this is what makes the loss of experience at a move between job types costly, and what the wage changes at exits from overeducation discipline in the next step.

The wage data identify output over the range of experience observed between ages 30 and 55, but not its level at near-zero experience: along the direction in which $\phi_{1,f}$ rises and ρ_f falls, the fit is almost unchanged. I therefore impose that a college job without experience produces at least as much as a non-college job without experience, $\rho_f \geq \rho_0$, a restriction that binds for business/law and STEM jobs. Without it, the fit improves slightly (the objective falls from 0.467 to 0.439), output at zero experience in these jobs falls to almost nothing, but the returns to experience over the observed range are almost unchanged (46 and 47 log points between 5 and 20 years), and so are the Step 2 estimates (Section 4.4).

4.4 Step 2: Search, Learning and Mobility Parameters

The 20 remaining parameters are the exogenous separation rates δ_0 and δ_c, the search probability of employed workers λ_e, the learning probability α, the vacancy costs $k_0, \ldots, k_3$, the probabilities of a good match $\bar{a}_0, \ldots, \bar{a}_3$, the growth of the flow value of unemployment with age g_b, the probability of an exogenous offer of a matched job π, common to all fields, the shares of experience kept at a move between job types τ_1, τ_2, τ_3, and the amenities of good matches in college jobs χ_1, χ_2, χ_3. They are estimated by minimum distance, a GMM estimator whose moment conditions are the differences between model and data moments. Let $\hat{m}_k$ be the k-th of the $K = 42$ targets, $\hat{\sigma}_k$ its

Figure 5: Age profile of wages: model and data.



Note: Mean log real monthly wage by age. Dots: NLSY79 data, as in Figure 4. Dashed lines: model, the log wage of a good match (a constant share of its output g, with the parameters of Table 6), averaged over the ten deciles of field-specific experience observed in the data for each group and age. Left panel: matched workers, by field of the job (non-college workers in non-college jobs and graduates in college jobs). Right panel: overeducated graduates by field of study, whose experience is in non-college jobs. Wages are relative to the monthly federal minimum wage of 2000, as in Figure 4.

Table 6: Production-function parameters (Step 1).

Parameter	Description	Value
ρ_0	Baseline productivity: non-college jobs	1.142
ρ_1	Baseline productivity: humanities jobs	1.914
ρ_2	Baseline productivity: business jobs	1.142
ρ_3	Baseline productivity: STEM jobs	1.142
μ_1^{ov}	Overeducation premium: humanities graduates	0.501
μ_2^{ov}	Overeducation premium: business graduates	0.833
μ_3^{ov}	Overeducation premium: STEM graduates	0.831
$\phi_{1,0}$	Experience weight: non-college jobs	0.225
$\phi_{1,1}$	Experience weight: humanities jobs	0.214
$\phi_{1,2}$	Experience weight: business jobs	0.874
$\phi_{1,3}$	Experience weight: STEM jobs	0.935
ϕ_2	Concavity of the experience profile	0.326

Note: Parameters of the output of a match, $g = \left[\rho_{f_j} + \mu_{f_i}^{ov}\mathbf{1}\{f_i > 0, f_j = 0\}\right] h_{f_j}(e)$ with $h_{f_j}(e) = (1-\phi_{1,f_j}) + \phi_{1,f_j}(1+e)^{\phi_2}$, where f_i is the worker's field of study, f_j the field of the job and e the experience in the field of the job, in months. The baseline productivities of business/law and STEM jobs are at the lower bound $\rho_f \geq \rho_0$ (Section 4.3). The parameters minimize the sum of squared deviations between the mean log wages of the model and of the data for matched workers by field of the job and overeducated graduates by field of study, at each age between 30 and 55 (182 moments, Figure 5).

standard error, clustered by worker, and $m_k(\Theta)$ its model counterpart. The parameters solve

$$\hat{\Theta} = \arg\min_{\Theta} \frac{1}{K}\sum_{k=1}^{K}\left(\frac{m_k(\Theta) - \hat{m}_k}{\hat{\sigma}_k}\right)^2, \tag{14}$$

so that the weighting matrix is diagonal, with the inverse of the sampling variance of each target on the diagonal, and the objective is the average squared t-statistic of the differences between model and data. The diagonal weights make targets measured in different units and with different precision comparable. I do not use the efficient weighting matrix, the inverse of the full covariance matrix of the targets, whose estimation biases the estimates in finite samples (Altonji and Segal, 1996). The targets are:

- the UE rate of non-graduates by age band (3 moments);
- the UM rate of graduates by field and age band (9);
- the OM rate of overeducated graduates by field and age band (9);
- the OU rate of overeducated graduates by age band (3);
- the rate of moves into unemployment by job tenure (1–12, 13–36 and 37 or more months) for non-college jobs of non-graduates (3) and of overeducated graduates (3);
- the separation rate by job tenure for college jobs of each field (9);
- the change in the log wage at transitions from overeducation to a matched job, relative to overeducated stayers, by field (3; Table 5).

Model moments are computed exactly, without simulation, from the policy functions at the empirical distribution of workers across fields, experience and job tenure at each age, the latter determining the share of matches whose quality is still unknown. In college jobs, the model counterpart of the separation rate is the rate of moves into unemployment, since matched workers do not move to other college jobs in the model. Similarly, for matched workers in non-college jobs the targets are the moves into unemployment, since the model has no moves between non-college employers. The model counterpart of the wage change at a transition is the change in the log output of a good match between the non-college job and the matched job with the share τ_{f_i} of experience, averaged over the transitions predicted by the model. Under the wage convention of Section 3, this is the change in the wage even though the new match is of unknown quality; Appendix C shows that, if wages were instead a share of the expected output, the model would predict large wage losses at these transitions for any transferability of experience.

Identification. Table 7 summarizes which moments discipline each channel. The vacancy costs are pinned down by the job-finding rates of the unemployed: k_0 by the UE rate of non-graduates and k_f by the UM rate of graduates of field f. The tenure

profiles of separations discipline learning: the decline of separations over the first years of tenure identifies the speed of learning α, and the level of early-tenure separations the probability of a good match $\bar{a}_f$, while separations at long tenure pin down the exogenous separation rates. Given the job-finding rates, the OM rates by field and age band identify the intensity of on-the-job search λ_e, the exogenous offers π and the amenities χ_f. The offers shift the OM rates of all fields, and their probability is pinned down by the level of the OM rates common to the three fields, in particular at older ages, when few workers search. The amenity χ_f raises the value of a good match in a college job of field f without changing wages: it raises the rates at which graduates of that field find matched jobs, and, since a match of unknown quality yields the amenity in expectation, it lowers the separations from these jobs in the first years of tenure. The share of experience kept at a move, τ_f, is disciplined by the wage changes at exits from overeducation: given the experience profile of wages estimated in Step 1, a low τ_f implies wage losses at these transitions, while the data show gains. It also affects the age profile of OM rates, since a low τ_f discourages moves more for experienced workers. Finally, the growth of the flow value of unemployment g_b is identified by the decline of the UE rate of non-graduates after age 50.

Table 7: Channels of persistence: parameters and identifying moments.

Channel	Parameters	Main identifying moments
Search frictions	k_0, k_1, k_2, k_3	UE rate of non-graduates; UM rate of graduates by field and age
Learning frictions	α, $\bar{a}_0, \ldots, \bar{a}_3$	Separation and EU rates by job tenure
Loss of experience	τ_1, τ_2, τ_3	Wage change at transitions from overeducation; OM rate by age
Aging (horizon)	T	Retirement age
On-the-job search	λ_e	OM and OU rates
Exogenous offers	π	OM rate at 50–55
Amenities	χ_1, χ_2, χ_3	Separation rates by job tenure; UM and OM rates by field
Unemployment flow value growth	g_b	UE rate of non-graduates at 50–55
Separations	δ_0, δ_c	EU and separation rates at long tenure

Note: For each channel of persistence and each other component of the model, the parameters and the targets that are most informative about them. The Step 2 parameters are estimated jointly, so every target depends on all of them; Appendix Table A2 reports the sensitivity of each estimate to the targets. Monthly transition rates: UE, from unemployment to employment; UM, from unemployment to a matched job; OM, from overeducation to a matched job; EU, from a non-college job to unemployment. Separation rate: monthly rate of leaving a college job. Job tenure is grouped in 1–12, 13–36 and 37 or more months. The parameters are described in Tables 6 and 8; the retirement age T (65) is assigned.

Calibrated parameters and fit. Table 8 reports the calibrated parameters and Figure 6 the fit of the targets; Appendix Table A1 lists every target. Overall, the

model reproduces the targets quite well: the average squared t-statistic is 1.6, and for 37 of the 42 targets the model moment falls within the 95% confidence interval of the data. The model misses five targets, all of them separation rates by job tenure. The largest miss concerns the separation rate of new matches in humanities jobs in the first year of tenure, which the model underpredicts by about a quarter (3.9% per month against 5.2% in the data). The other four misses are smaller, between 0.08 and 0.7 percentage points per month and 15–25% of the level of the data, and lie just outside the confidence interval (2.0 to 2.9 standard errors): the model overpredicts separations from humanities jobs in the second and third years of tenure and from business/law and STEM jobs in the first year, and the rate at which overeducated graduates with more than three years of tenure lose their job. All the flows into and out of overeducation, and the wage changes at exits from overeducation, are matched within their confidence intervals.

The learning probability $\alpha = 0.063$ implies that the quality of a match is learned on average after 16 months, and that it is still unknown after one year in 46% of matches; a new match is good with probability 0.62 in non-college jobs and 0.41–0.50 in college jobs, the lowest in humanities. The vacancy costs of business/law and STEM jobs are about six and a half times those of non-college jobs, and those of humanities jobs about three times. Employed workers search with probability $\lambda_e = 0.66$. Graduates keep 43–48% of their experience when moving between non-college and business/law or STEM jobs, and 87% of it in humanities ($\tau_1 = 0.87$), where experience matters less for wages. The exogenous offers of a matched job arrive with a monthly probability of 0.27%. The amenity of a good humanities job, $\chi_1 = 1.02$, is large, about 27% of the average output of a good humanities match at ages 30–55: without it, overeducated humanities graduates would have little reason to search for a matched job, since the wage premium of humanities jobs over non-college jobs is small and does not grow with experience. The amenity of business/law jobs is small, about 5% of output, and that of STEM jobs is zero. These estimates do not depend on the normalization of Step 1: with the production function estimated without the restriction $\rho_f \geq \rho_0$, the fit of Step 2 is similar (average squared t-statistic 1.7) and so are the parameters, for instance the shares of experience kept at a move ($\tau_f = 0.88$, 0.49 and 0.44).

Why humanities graduates differ. The weaker fit for humanities graduates, and in particular the slow decline of their OM rate between the thirties and the forties, follows from how the data discipline the channels of persistence in this field. In the data, the OM rate falls between the thirties and the forties by a similar proportion in all fields: by 58% for humanities, 60% for business/law and 68% for STEM graduates. In the model, three forces make exits from overeducation fall with age. First, the loss

of experience at a move becomes more costly as non-college experience accumulates, when experience is only partly kept and is rewarded more in college jobs. Second, the horizon over which the gain from a matched job is enjoyed shortens. Third, older overeducated workers are more often in matches whose good quality is already known. For business/law and STEM graduates, the first force accounts for a large part of the decline. For humanities graduates, the data shut it down: the wage premium of humanities jobs over non-college jobs is small and does not grow with experience, so that in Step 1 the output of a good match rises by 25 log points between 5 and 20 years of experience in humanities jobs, against 26 in non-college jobs and 44 in business/law and STEM jobs; and the wage gains at exits from overeducation imply that experience is largely kept ($\tau_1 = 0.87$).[5] The model must then generate the decline of humanities exits from the horizon and from learning alone, which change smoothly with age, and it predicts a decline of 35% instead of 58% between the thirties and the forties. The miss therefore points to an age-related force common to all fields which, for humanities graduates, the loss of experience cannot supply. It may be that part of the decline in the data comes from differences in ability among humanities graduates (the model abstracts from these differences), as discussed in Section 2.3.

5 Validation

This section compares the calibrated model with the two stylized facts presented in Section 2.3: the rise of persistence with age (Fact 1), which is the main object of the paper, and the survival of overeducation in the cohort of graduates overeducated at age 30 (Fact 2).

Persistence by age. Figure 8 compares the monthly OO, OM and OU rates of overeducated graduates by age with their empirical counterparts. The average OM rates by field over the three age bands and the average OU rates over the same bands are targeted, but their profiles within the bands and the OO rate are not. The model reproduces the rise of persistence with age: the OO rate increases from 0.96 at age 30

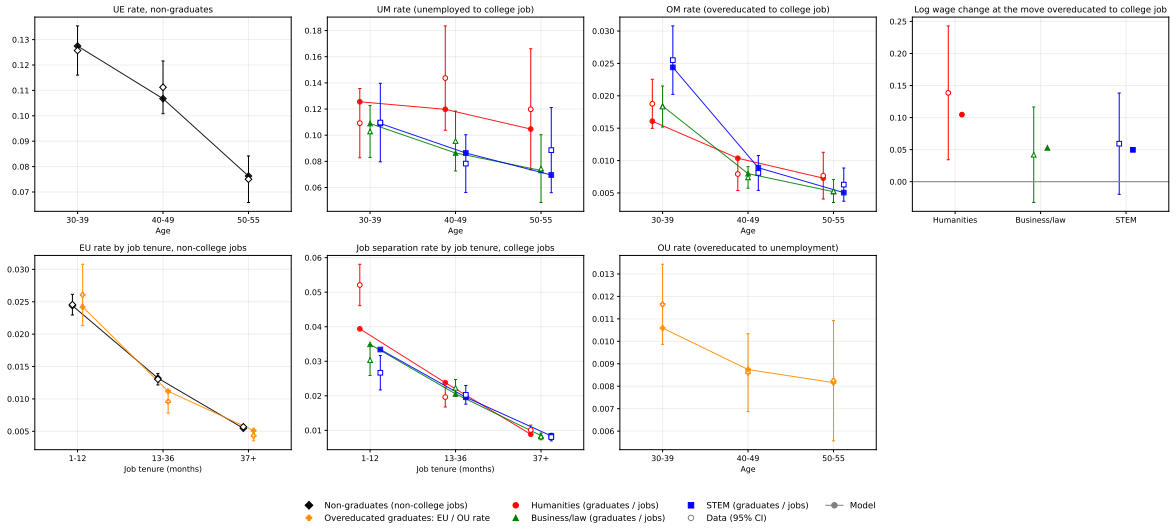
[5] In Step 1, the college jobs of humanities graduates include those in business/law or STEM occupations, 37% of their months in college jobs. This pooling of horizontal mismatch does not drive these results. Humanities graduates gain about as much from experience in humanities occupations as in other college occupations (42 and 47 log points between 5 and 20 years of experience), although they earn less in the former (mean log wage 1.18 against 1.54). Their wage gain at exits from overeducation is larger when they move into a humanities occupation (0.18, s.e. 0.09) than into another college occupation (0.09, s.e. 0.04), so the high value of τ_1 does not reflect moves into better-paid fields. Finally, their OM rate falls with age in the same proportion for both destinations, from 0.99% to 0.40% per month into humanities occupations and from 0.89% to 0.38% into other college occupations between ages 30–39 and 40–55. Restricting the wages of humanities jobs to humanities occupations would therefore lower their premium over non-college jobs without steepening their experience profile.

Table 8: Search, learning and mobility parameters (Step 2).

Parameter	Description	Value	(s.e.)	Main target
δ_0	Exogenous separation: non-college jobs	0.0049	(0.0002)	EU rate at long tenure, non-college jobs
δ_c	Exogenous separation: college jobs	0.0074	(0.0004)	Separation rate at long tenure, college jobs
λ_e	Search intensity while employed	0.662	(0.173)	OM and OU rates
α	Learning probability	0.063	(0.003)	EU and separation rates by job tenure
k_0	Vacancy cost: non-college jobs	39.38	(2.64)	UE rate of non-graduates
k_1	Vacancy cost: humanities jobs	121.51	(22.06)	UM rate, humanities
k_2	Vacancy cost: business jobs	253.35	(34.80)	UM rate, business/law
k_3	Vacancy cost: STEM jobs	262.96	(31.70)	Separation and UM rates, STEM
$\bar{a}_0$	Prob. good match: non-college jobs	0.616	(0.014)	Early-tenure EU rate, non-college jobs
$\bar{a}_1$	Prob. good match: humanities jobs	0.410	(0.026)	Early-tenure separation rate, humanities jobs
$\bar{a}_2$	Prob. good match: business jobs	0.480	(0.017)	Early-tenure separation rate, business jobs
$\bar{a}_3$	Prob. good match: STEM jobs	0.505	(0.030)	Early-tenure separation rate, STEM jobs
g_b	Growth of b with age (per year)	0.0202	(0.0008)	UE rate by age, non-graduates
π	Offer of a matched job to the overeducated (all fields)	0.003	(0.001)	OM rate at 50–55
τ_1	Experience kept moving between job types: humanities	0.873	(0.168)	Wage change at OM moves, humanities
τ_2	Experience kept moving between job types: business	0.476	(0.042)	Wage change at OM moves, business/law
τ_3	Experience kept moving between job types: STEM	0.425	(0.041)	Wage change at OM moves, STEM
χ_1	Amenity of a good match: humanities jobs	1.015	(0.222)	Separation and UM rates, humanities
χ_2	Amenity of a good match: business jobs	0.304	(0.140)	UM and OM rates, business/law
χ_3	Amenity of a good match: STEM jobs	0.009	(0.145)	Separation and OM rates, STEM

Note: Minimum-distance estimates with a diagonal weighting matrix (the inverse of the squared standard errors of the 42 targets, clustered by worker). The model period is one month: δ_0, δ_c, λ_e, α and π are monthly probabilities. Vacancy costs k_f and amenities χ_f (a monthly flow enjoyed in a good match of a college job of field f, not paid as a wage) are in units of output; τ_f is the share of experience kept when moving between a non-college and a college job; the flow value of unemployment at age t is $B_{30}\left[1 + g_b(t-30)\right]$. "Main target": the targets that contribute most to the precision of each estimate (Appendix Table A2; all targets are listed in Appendix Table A1); all parameters are estimated jointly. Standard errors in parentheses are the square roots of the diagonal of $(G'WG)^{-1}$, where G is the Jacobian of the model moments with respect to the parameters and $W = \text{diag}(1/\hat{\sigma}_k^2)$; G is computed by central differences with steps of 5% of each parameter, since the model moments change in small discrete jumps with the parameters (the policies are chosen on a discrete grid of experience). The standard errors ignore the covariance between targets and the estimation error of the Step 1 parameters. Assigned parameters: $\beta = 0.9967$, $\lambda_u = 1$, B_{30} equal to 71% of the output of a good non-college match of non-graduates at age 30.

Figure 6: Calibration targets: model and data.



Note: The 42 targets of Step 2. Hollow markers with 95% confidence intervals: NLSY79 data (standard errors clustered by worker); filled markers joined by lines: calibrated model. Top row: monthly rates by age band of unemployed non-graduates finding a job (UE), of unemployed graduates finding a matched job (UM), and of overeducated graduates moving to a matched job (OM), by field; change in the log wage at OM transitions relative to overeducated stayers, by field (Table 5). Bottom row: monthly rates by job tenure of moves from non-college jobs to unemployment (EU), for non-graduates and overeducated graduates, and of separations from college jobs, by field of the job; monthly rate of overeducated graduates moving to unemployment (OU) by age band. Colors and markers denote the field of study of graduates or the field of the job, as in the legend.

to 0.99 at age 55, and the model accounts for 90% of the increase between the thirties and ages 50–55 in the data (1.6 percentage points against 1.75). The model rates lie within the 95% confidence interval of the data at 22 of the 26 ages for the OO and OM rates and at 24 for the OU rate. The model overpredicts persistence at ages 30–33, because it underpredicts the exits of young overeducated graduates both to matched jobs and to unemployment.

Survival of overeducation. Figure 7 simulates the cohort of graduates overeducated at age 30, starting from the empirical distribution of their experience and job tenure, and follows the exact distribution of the cohort over the states of the model. The model reproduces the fast initial exits: for business/law graduates, the share that has not yet held a matched job is 0.30 at age 35 in the model, against 0.34 in the data, and for STEM graduates 0.21, against 0.25. It does not reproduce the long flat tail of the survival curve for these graduates, however: the share falls to 0.11 at 45 and 0.06 at 55 for business/law graduates, against 0.23 and 0.19 in the data, and to 0.04 and 0.02 for STEM graduates, against 0.18 and 0.16. For humanities graduates, exits are slower in the first years (0.49 at age 35 against 0.38 in the data), while the share is close to the

data from the mid-forties on (0.20 at 45 and 0.09 at 55, against 0.18 and 0.12).

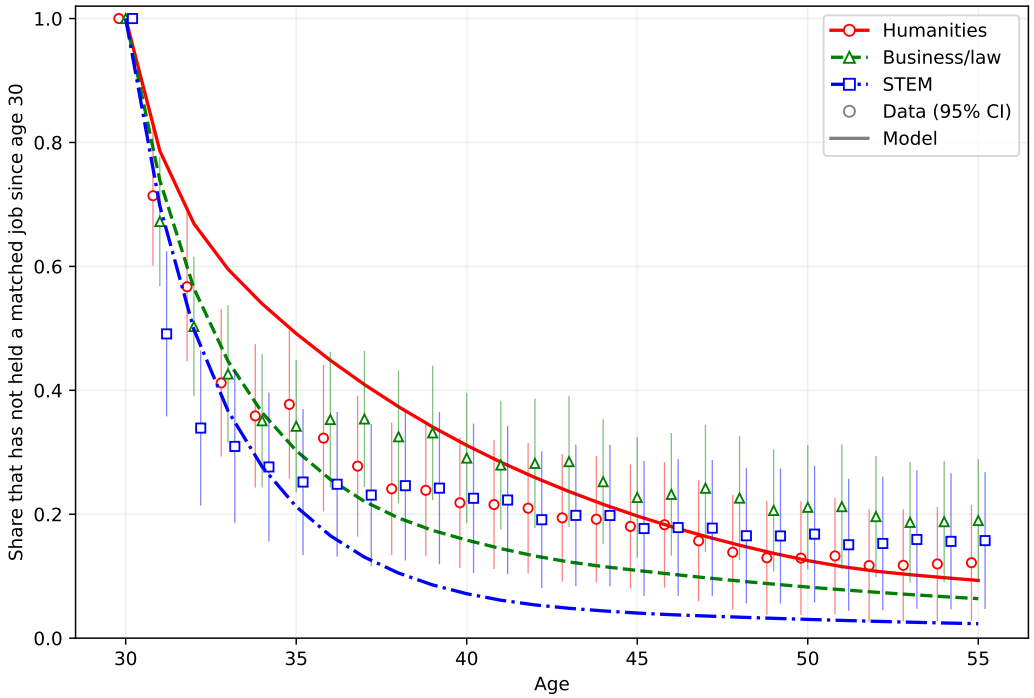
Why the tail is too short. The model and the data differ in who is overeducated later in the career. Most graduates who are overeducated after age 40 have held a college job earlier in the panel, and these returners leave overeducation much faster than graduates who have never held one (Table 9): at ages 40–49, the monthly OM rate is 0.88% for the former and 0.35% for the latter, and returners account for 80% of the overeducated months at those ages. In the model, matched graduates never move back to a non-college job, so the overeducated graduates of the model are mostly graduates who have not held a college job since 30, and the targeted OM rates, which in the data mix the two groups, are attributed to them. The model therefore generates the average exit rate from overeducation at each age, which drives the monthly persistence of Figure 8, but it overstates the exits of the graduates who have never held a college job, whose survival Figure 7 follows. The pattern is consistent with the mechanism of the model: returners have accumulated experience in college jobs, which makes a move back to a college job less costly. Capturing it requires tracking experience in college and non-college jobs separately and allowing moves from college jobs back into overeducation, an extension that I leave for future work. For the decomposition of Section 6, the implication is that the measures based on the cohort overeducated at 30 understate the persistence of business/law and STEM graduates, whereas the measures based on the monthly rates by age do not.

Table 9: Exits from overeducation by employment history.

	OM rate (%)		Share of returners in
	Never held a college job	Held a college job	overeducated months (%)
Ages 30–39	1.53	2.40	57
	(0.16)	(0.15)	
Ages 40–49	0.35	0.88	80
	(0.09)	(0.08)	
Ages 50–55	0.27	0.68	84
	(0.11)	(0.08)	

Note: NLSY79 college graduates overeducated (employed in a non-college occupation) in a month, ages 30–55. OM rate: share employed in a college occupation in the following month, in percent, for graduates who have not held a college occupation in any earlier month of the panel since age 30 and for those who have (returners); standard errors clustered by worker in parentheses. Last column: share of the overeducated months of returners. Weighted by the cross-sectional sampling weights.

Figure 7: Cohort of graduates overeducated at age 30: model and data.



Note: Graduates overeducated (employed in a non-college occupation) in their first month observed at age 30. At each age, share of the cohort that has not held a matched job (a college occupation) since age 30, by field of study, measured in the first month of that age. Hollow markers with 95% confidence intervals: NLSY79 data, as in Figure 1, weighted by the cross-sectional sampling weights. Lines: calibrated model. The model cohort starts from the experience and job tenure of each cohort member at age 30, and its distribution over the model states is moved forward month by month with the policy functions, without simulation noise. These shares are not targeted in the calibration.

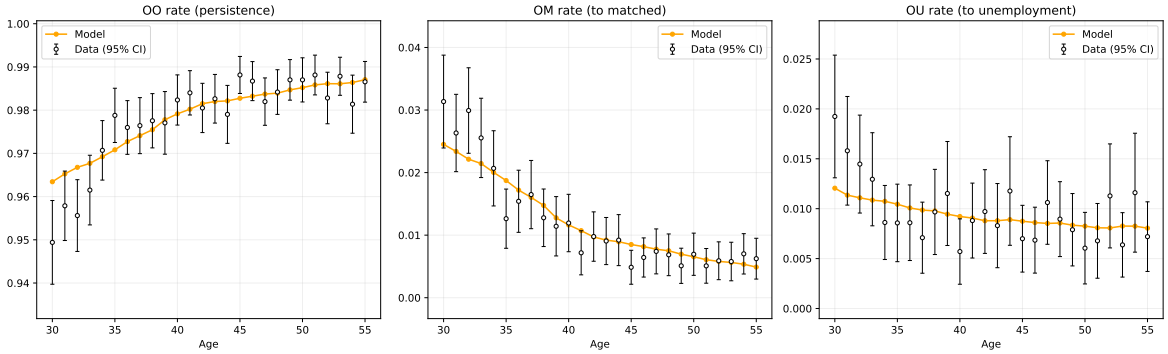
6 Decomposition of Overeducation Persistence

6.1 Methodology

I measure the contribution of each channel of Section 3.3 by reducing it and solving the model again, including the equilibrium tightness of every submarket. Each channel is reduced by a share s of its size, measured in its natural unit:

- *Search frictions*: the vacancy costs k_f are scaled by $1 - s$. For interior choices of tightness, this scales the expected duration of search by $1 - s$.
- *Loss of experience*: the share of experience lost at a move between job types, $1 - \tau_f$, is scaled by $1 - s$.
- *Learning frictions*: the expected time to learn the quality of a match, $1/\alpha$ months, is reduced by the share s of its distance from one month.

Figure 8: Monthly transition rates from overeducation by age: model and data.



Note: College graduates of all fields who are overeducated (employed in a non-college occupation) in a month. At each age, share that in the following month is still overeducated (OO, left), is employed in a matched job, i.e. a college occupation (OM, center), or is unemployed (OU, right). Hollow markers with 95% confidence intervals: NLSY79 data, weighted by the cross-sectional sampling weights (standard errors clustered by worker). Lines: calibrated model, computed at the empirical distribution of overeducated graduates across fields, experience and job tenure at each age. Only the average OM rates by field and age band (30–39, 40–49, 50–55) and the average OU rates by age band are calibration targets; the profiles within the bands and the OO rate are not targeted.

- *Aging, horizon*: the remaining work horizon at each age is increased by the share s of its distance from the horizon of a 30-year-old. The model is solved again for each age, with the value of unemployment of that age.

The exogenous offers π and the amenities χ_f are kept at their calibrated values. The offers stand for transitions that the model does not explain, such as internal reassignments and referrals, which workers rationally anticipate; removing them would change the environment rather than a friction, and the decomposition therefore measures the contribution of the four endogenous channels in the calibrated environment.

The main exercise halves every channel ($s = 0.5$). The channels interact, so the effect of halving one of them depends on the others. I therefore attribute the change in each measure of persistence with exact Shapley values over the $2^4 = 16$ economies in which each channel is either at its calibrated value or halved (Shorrocks, 2013). The contribution of a channel is its average marginal effect over all the orderings in which the channels can be halved, and the contributions add up to the difference between the calibrated economy and the economy with all the channels halved.

I consider five measures of persistence, all computed separately for humanities graduates and for business/law and STEM graduates pooled. Three are computed at the empirical distribution of experience and job tenure of overeducated graduates at each age, so that they measure the change in the behavior of a given population of overeducated workers: the average monthly OO rate at ages 30–39, the average monthly OO rate at ages 50–55, and the age gradient of persistence, the difference between the two.

These are the measures that the model reproduces best (Section 5). The other two are computed on the cohort of graduates overeducated at age 30, simulated as in Section 5, so that the distribution of workers across states responds to the counterfactual: the expected number of years spent in overeducation between ages 30 and 55, and the share that has never held a matched job by age 55. Since the model understates the long tail of the survival curve of business/law and STEM graduates, the cohort measures understate the persistence of these graduates, and their decomposition describes the persistence that the model generates. Finally, the model has a single worker type in each field, so the vacancy costs and the probabilities of a good match also absorb the differences in ability among graduates of the same field: the shares of the levels of persistence attributed to search and learning frictions include the lower exit rates of low-ability graduates (Section 2.3). The age gradient does not, since ability composition accounts for almost none of the decline of exits with age.

6.2 Results

Table 10 and Figure 9 report the Shapley decomposition. The level of persistence and its rise with age have different sources.

The rise of persistence with age. In the calibrated economy, the monthly OO rate of overeducated business/law and STEM graduates is 1.78 percentage points higher at ages 50–55 than in their thirties, and the gradient would turn negative (−0.46 points) if all the channels were halved. The loss of experience (+1.49 points) and the shrinking horizon (+1.10 points) account for the gradient, learning frictions contribute marginally (+0.16 points), and search frictions reduce it (−0.51 points): frictions raise persistence more for young workers, who have more to gain from a matched job, and flatten the persistence profile. The decomposition of the OO rate by age band makes the contrast explicit. In the thirties, persistence is due almost entirely to search frictions (76%) and learning frictions (29%), and the loss of experience slightly lowers it (−8%): for young graduates, the more experience they stand to lose, the higher the cost of remaining overeducated. At ages 50–55, search and learning frictions account for 36% and 20% of persistence, and the loss of experience (22%) and the horizon (22%) together for more than two fifths. The loss of experience thus works in two directions. Young overeducated graduates, who have accumulated little non-college experience, anticipate the cost of remaining overeducated and are more likely to leave early; those who remain overeducated accumulate experience in non-college jobs and become increasingly unlikely to leave because the cost of leaving becomes higher. The age gradient of persistence is therefore the signature of this specialization trap and of the shrinking work horizon.

The level of persistence. For business/law and STEM graduates overeducated at 30, the expected time spent in overeducation between ages 30 and 55 is 4.4 years in the calibrated economy, and 1.5 years when all channels are halved. Search frictions account for 59% of the difference, learning frictions for 33% and the horizon for 8%. The loss of experience does not contribute (0%), because its two effects offset each other: it accelerates the early exits, which make up most of the time spent in overeducation by this cohort, and it lengthens the spells of those who remain. The second effect dominates the share of the cohort that never reaches a matched job by age 55, 5% in the model, which would fall to zero: it is attributed in almost equal parts to search frictions, learning frictions and the loss of experience (26% each), and for 22% to the horizon.

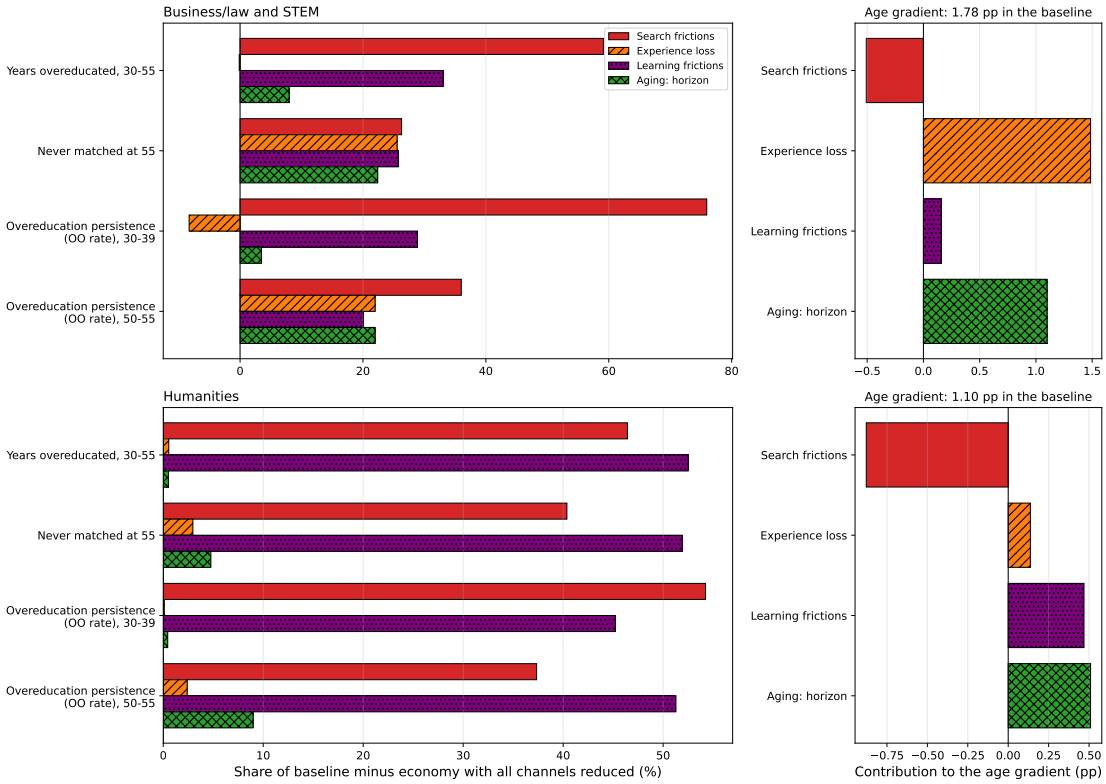
Table 10: Shapley decomposition of persistence: contribution of each channel, channels halved.

	Business/law and STEM					Humanities				
	Years	Never at 55	OO 30–39	OO 50–55	Gradient	Years	Never at 55	OO 30–39	OO 50–55	Gradient
Search frictions	59%	26%	76%	36%	-0.51 pp	46%	40%	54%	37%	-0.88 pp
Experience loss	-0%	26%	-8%	22%	+1.49 pp	1%	3%	0%	2%	+0.14 pp
Learning frictions	33%	26%	29%	20%	+0.16 pp	53%	52%	45%	51%	+0.47 pp
Aging: horizon	8%	22%	3%	22%	+1.10 pp	1%	5%	0%	9%	+0.51 pp
Baseline	4.4	0.047	0.969	0.987	1.78 pp	7.7	0.093	0.973	0.985	1.10 pp
All channels halved	1.5	0.000	0.936	0.931	-0.46 pp	1.1	0.000	0.916	0.925	0.87 pp

Note: Each channel is halved as described in Section 6.1. The last two rows report each measure of persistence in the calibrated economy ("Baseline") and in the economy in which all four channels are halved. The rows above attribute the difference between the two to each channel with exact Shapley values over the 16 economies in which each channel is either at its calibrated value or halved: as a share of the difference (shares sum to 100%), and for the gradient in percentage points (contributions sum to the difference). A negative entry means that halving the channel raises persistence. Measures, for business/law and STEM graduates pooled and for humanities graduates: Years, expected years spent in overeducation between ages 30 and 55, and Never at 55, share that has never held a matched job by age 55, both for the cohort of graduates overeducated at age 30; OO 30–39 and OO 50–55, average monthly probability of remaining overeducated at those ages, computed at the empirical distribution of experience and job tenure at each age; Gradient, OO 50–55 minus OO 30–39.

Marginal effects. Table 11 reports the effect of reducing each channel by 10% alone ($s = 0.1$), and Figure 10 the effect of reducing each channel from 0 to 100%. At the margin, the three main channels have similar effects on business/law and STEM graduates: a 10% reduction of search frictions lowers the time spent in overeducation between 30 and 55 by 1.0 year, a 10% reduction of learning frictions by 0.9 years, and a 10% reduction of the loss of experience by 0.6 years, and each lowers the share that never reaches a matched job by 2.2–2.4 percentage points. The effect of the horizon is smaller (0.2 years). Figure 10 shows that the effect of search frictions keeps growing up to their complete removal, and that of learning frictions levels off once half of them is removed. The time spent in overeducation falls with the first quarter of the loss of

Figure 9: Shapley decomposition of persistence, channels halved.



Note: Shapley decomposition of Table 10, for business/law and STEM graduates pooled (top) and humanities graduates (bottom). Left panels: share of the difference between the calibrated economy and the economy with all four channels halved that is attributed to each channel, for four measures of persistence: years spent in overeducation between ages 30 and 55 and share never matched by age 55 (cohort of graduates overeducated at age 30), and average monthly OO rate at ages 30–39 and 50–55. For each measure the shares sum to 100%; a negative share means that halving the channel raises persistence. Right panels: contribution of each channel to the age gradient of persistence (OO rate at 50–55 minus OO rate at 30–39), in percentage points; the panel title reports the gradient in the calibrated economy.

experience removed and rises again afterwards, while the share never matched keeps falling: the two effects of the loss of experience described above.

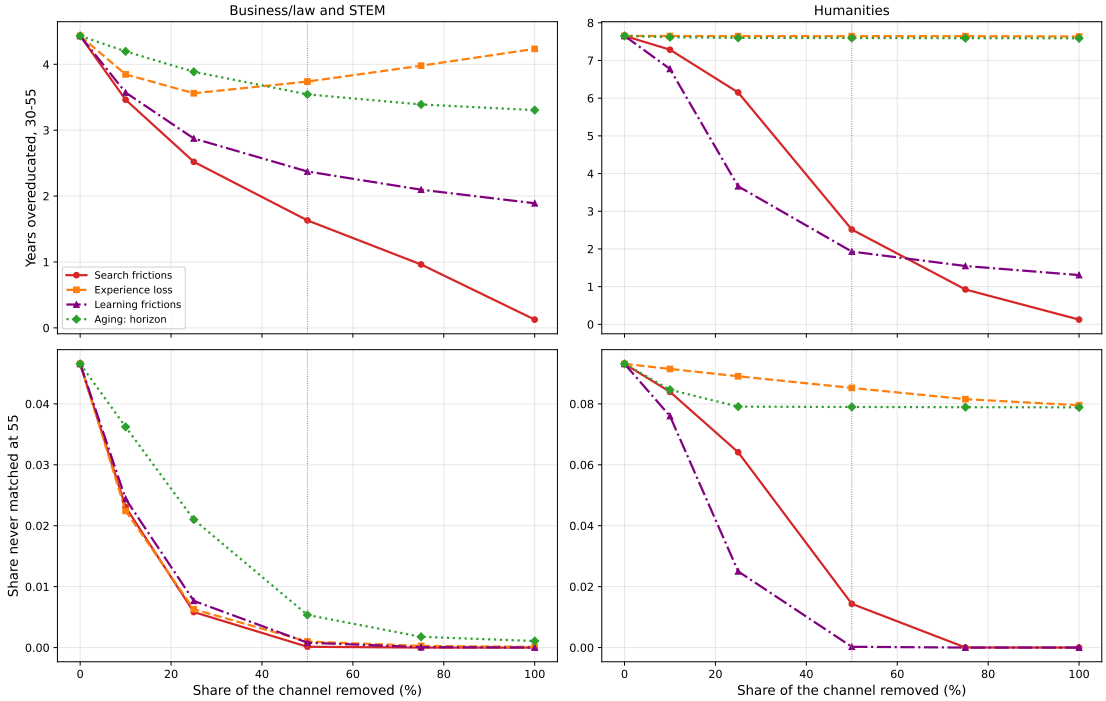
Humanities graduates. Humanities graduates overeducated at 30 spend 7.7 years in overeducation between ages 30 and 55 in the calibrated economy, and 1.1 years when all channels are halved. Learning frictions account for 53% of the difference and search frictions for 46%, and they account for 52% and 40% of the share that never reaches a matched job. The loss of experience plays no role, since humanities graduates keep most of their experience at a move ($\tau_1 = 0.87$). The weight of learning reflects the low probability that a new humanities match is good ($\bar{a}_1 = 0.41$): overeducated humanities graduates search for a matched job mainly while the quality of their current match is unknown, and the speed of learning governs how long they remain in that state.

Table 11: Effect of a 10% reduction of each channel.

	Business/law and STEM				Humanities			
	Years	Never at 55 (pp)	OO 50–55 (pp)	Gradient (pp)	Years	Never at 55 (pp)	OO 50–55 (pp)	Gradient (pp)
Search frictions	-0.97	-2.35	-0.051	+0.370	-0.37	-0.92	-0.062	+0.169
Experience loss	-0.58	-2.41	-0.084	+0.015	-0.00	-0.17	-0.010	-0.007
Learning frictions	-0.86	-2.22	-0.016	+0.279	-0.88	-1.71	-0.059	+0.271
Aging: horizon	-0.24	-1.03	-0.090	-0.022	-0.03	-0.85	-0.063	-0.058
Baseline	4.43	4.66	98.67	1.78	7.65	9.31	98.45	1.10

Note: Each cell is the change in a measure of persistence when a single channel is reduced by 10% ($s = 0.1$, Section 6.1) and the other channels are kept at their calibrated values, relative to the calibrated economy ("Baseline", last row). A negative entry means that reducing the channel lowers persistence. Measures, for business/law and STEM graduates pooled and for humanities graduates: Years, expected years spent in overeducation between ages 30 and 55, and Never at 55, share that has never held a matched job by age 55, both for the cohort of graduates overeducated at age 30; OO 50–55, average monthly probability of remaining overeducated at ages 50–55; Gradient, OO rate at 50–55 minus OO rate at 30–39, both computed at the empirical distribution of experience and job tenure at each age. Changes in years for Years and in percentage points otherwise; in the last row, years and percentages.

Figure 10: Persistence as each channel is reduced.



Note: Each line reduces a single channel by the share s on the horizontal axis (Section 6.1), from the calibrated economy ($s = 0$) to the complete removal of the channel ($s = 100\%$), keeping the other channels at their calibrated values; markers at $s = 10$, 25, 50, 75 and 100%. Top panels: expected years spent in overeducation between ages 30 and 55; bottom panels: share that has never held a matched job by age 55. Both measures refer to the cohort of graduates overeducated at age 30, for business/law and STEM graduates pooled (left) and humanities graduates (right). The vertical dotted line marks $s = 50\%$, the halving used in the Shapley decomposition of Table 10.

The age gradient of persistence, 1.10 percentage points, is raised by the horizon (+0.51 points) and by learning frictions (+0.47 points) and reduced by search frictions (−0.88

points), so that halving all channels lowers it only to 0.87 points.

7 Conclusion

For a sizable minority of U.S. college graduates, overeducation is not a temporary stage of the career: the monthly persistence of overeducation rises with age, and among graduates overeducated at 30, one in six has not reached a job that requires a degree by 55. This paper has studied these facts with a lifecycle model of directed search in which graduates accumulate field-specific experience that is only partly transferable between non-college and college jobs, learn about the quality of their matches over time, and face a shrinking work horizon as they age. Disciplined by labor market flows, by the tenure profiles of separations and by the wage changes at exits from overeducation, the model reproduces the untargeted rise of persistence with age and the speed of the early exits from overeducation.

The decomposition shows that the level and the lifecycle profile of persistence have different sources. Search and learning frictions account for most of the time that graduates spend in overeducation, while the partial loss of experience and the shrinking work horizon account for the rise of persistence with age. The loss of experience works in two directions: it pushes young overeducated graduates to leave before their non-college experience accumulates, and it traps those who remain. Policies that reduce search and learning frictions shorten overeducation most when they act early in the career, when these frictions account for almost all of its persistence. Policies that preserve the value of experience across job types act on a different margin: they lower the persistence of older graduates and the share that never reaches a matched job, but they also weaken the incentive of young graduates to leave early, so that their effect on the total time spent in overeducation is small.

The model leaves out one feature of the data that matters for the longest spells: most graduates who are overeducated later in the career have held a college job before, and they leave overeducation much faster than those who never have. Allowing workers to move from college jobs back into overeducation while keeping track of their experience in both types of jobs would let the model separate these two groups, and is a natural direction for future work.

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Appendix

A Existence, Uniqueness and Block Recursivity of the Equilibrium

The proof adapts the argument in Theorem 1 of Menzio et al. (2016) to accommodate the additional heterogeneity introduced in this paper: field-specific vacancy costs $k_{f'_j}$, field-specific experience that is partly transferable between job types, the exogenous offers of a matched job that overeducated workers receive with probability π, the amenities χ_{f_j} of good matches in college jobs, the feasible job fields $\mathcal{F}(f_i)$, and match-quality priors and separation rates that depend on the field of the job. The strategy is backward induction on the finite horizon $t \in \{0, 1, \ldots, T\}$.

Result. A *recursive equilibrium* is a collection of value functions $\{U_t, V_t, R_t, S_t\}$, market tightness $\{\theta_t\}$, and policy functions (search submarket, separation probability and acceptance of exogenous offers) that satisfy the Bellman equations and the zero-profit condition of Section 3.2, together with the law of motion of the aggregate state ψ, the distribution of workers across individual states and ages, implied by the policy functions. I show that:

(i) in every recursive equilibrium, U_t, V_t, R_t, S_t and θ_t do not depend on ψ, and neither do the sets of optimal search, separation and acceptance choices;

(ii) these functions are the same in every recursive equilibrium;

(iii) a recursive equilibrium exists and, under the tie-breaking rule below, its policy functions are unique and do not depend on ψ. The equilibrium is therefore block recursive and unique.

The tie-breaking rule is the one used in the numerical solution: (a) a match is dissolved ($d = 1$) whenever the worker is indifferent between unemployment and the match, $U_t = V_t + \lambda_e S_t$; (b) a worker for whom no field gives a strictly positive expected gain from search does not search ($\theta = 0$), since otherwise every submarket with zero tightness would be optimal; (c) a worker who is indifferent between the two fields in $\mathcal{F}(f_i)$ searches in the non-college field; (d) an overeducated worker who is indifferent rejects the offer of a matched job.

Notation. Write $\boldsymbol{\omega} = (a, f_i, f_j, e)$, with $f_j \in \mathcal{F}(f_i)$, e the experience in field f_j, and age t kept separate; $\boldsymbol{\omega}(a)$ emphasizes the match quality. The flow value of unemployment and the search problem of an unemployed worker do not depend on a, so

$U_t(\boldsymbol{\omega}(a)) = U_t(\boldsymbol{\omega}(a^0))$ and $R_t(\boldsymbol{\omega}(a)) = R_t(\boldsymbol{\omega}(a^0))$. The state of a continuing match at the beginning of the next period is $\hat{\boldsymbol{\omega}}(a) = (a, f_i, f_j, e+1)$. A worker who searches from state $\boldsymbol{\omega}$ in field $f_j' \in \mathcal{F}(f_i)$ forms a new match in state

$$\nu(\boldsymbol{\omega}, f_j') = \big(a^0, f_i, f_j', \varepsilon(e, f_j, f_j')\big), \qquad \varepsilon(e, f_j, f_j') = e\,\mathbf{1}[f_j' = f_j] + \lfloor \tau_{f_i} e \rfloor\,\mathbf{1}[f_j' \neq f_j],$$

whose value is $\tilde{V}_t(\boldsymbol{\omega}, f_j') = V_t\big(\nu(\boldsymbol{\omega}, f_j')\big)$. Hence $\tilde{V}_t$ is a function of V_t alone. Finally, let $q(0) \equiv \lim_{\theta \downarrow 0} p(\theta)/\theta = p'(0^+) \in (0, \infty]$, with the convention $k/q(0) = 0$ if $q(0) = \infty$.

Lemma A.1 (free entry). Fix $k > 0$ and $\tilde{V} \in \mathbb{R}$. For every x, the conditions $k \geq q(\theta)\,[\tilde{V} - x]$, $\theta \geq 0$, with complementary slackness, have the unique solution

$$\theta(x) = \begin{cases} q^{-1}\left(\dfrac{k}{\tilde{V} - x}\right) & \text{if} \quad x < \tilde{V} - k/q(0), \\ 0 & \text{otherwise.} \end{cases}$$

Proof. The function q is continuous and strictly decreasing on $(0, \infty)$: on $(0, \bar{\theta}]$ because p is strictly concave with $p(0) = 0$, so that its average $p(\theta)/\theta$ is strictly decreasing, and on $[\bar{\theta}, \infty)$ because $q(\theta) = 1/\theta$ there. Since $q(0^+) = q(0)$ and $q(\infty) = 0$, q maps $(0, \infty)$ onto $(0, q(0))$ one-to-one. If $x < \tilde{V} - k/q(0)$, then $k/(\tilde{V} - x) \in (0, q(0))$, so the equality $k = q(\theta)[\tilde{V} - x]$ has exactly one solution $\theta > 0$, while $\theta = 0$ violates the inequality because $q(0)[\tilde{V} - x] > k$. If instead $x \geq \tilde{V} - k/q(0)$, every $\theta > 0$ gives $q(\theta)[\tilde{V} - x] < k$, so the equality cannot hold and $\theta = 0$ is the only solution. $\square$

Lemma A.2 (search). Fix $k > 0$, $\tilde{V} \in \mathbb{R}$ and an outside option $O \in \mathbb{R}$, and let $D = \tilde{V} - O$. With $\theta(x)$ given by Lemma A.1,

$$\max_x\, p(\theta(x))\,[x - O] = \max_{\theta \geq 0}\, \big\{ -k\,\theta + p(\theta)\,D \big\} \equiv \Pi(D; k).$$

The right-hand side has a unique maximizer $\theta^*(D; k) \in [0, \bar{\theta}]$, with $\theta^* > 0$ if and only if $D > k/q(0)$, and $\Pi(D; k) \geq 0$. For the calibrated $p(\theta) = \min\{\theta^{1/2}, 1\}$, $\bar{\theta} = 1$ and $\theta^*(D; k) = \min\{(D/2k)^2,\, 1\}$ if $D > 0$, and $\theta^*(D; k) = 0$ otherwise.

Proof. By Lemma A.1, the offers $x < \tilde{V} - k/q(0)$ are in one-to-one correspondence with $\theta \in (0, \infty)$ through $x = \tilde{V} - k/q(\theta)$, and for them $p(\theta)[x - O] = p(\theta)D - k\theta$, using $p(\theta)/q(\theta) = \theta$. All other offers have $\theta = 0$ and yield $p(0) = 0$, the value of the right-hand side at $\theta = 0$. Hence the two maxima coincide. If $D \leq 0$, the objective is at most $-k\theta$ and is maximized only at $\theta = 0$. If $D > 0$, the objective is negative for $\theta > D/k$, so a maximizer exists on the compact set $[0, D/k]$; it is unique because the objective is strictly concave on $[0, \bar{\theta}]$ and strictly decreasing on $[\bar{\theta}, \infty)$, where $p = 1$.

Its right derivative at zero is $-k + q(0)D$, which gives the condition for $\theta^* > 0$. For the calibrated p, $q(0) = \infty$; the first-order condition $D/(2\theta^{1/2}) = k$ on $(0,1)$ gives $\theta = (D/2k)^2$, and the solution is the corner $\bar{\theta} = 1$ when $D \geq 2k$. $\square$

Step 1: terminal period T. In the last period, the value of unemployment and the joint value of a match with known quality are

$$U_T(\boldsymbol{\omega}, \psi) = b_T, \qquad V_T(\boldsymbol{\omega}(a), \psi) = z(a, f_i, f_j, e), \quad a \in \{a^G, a^B\}.$$

With no continuation value, the joint value of a match with unknown quality is

$$V_T(\boldsymbol{\omega}(a^0), \psi) = \alpha\, \mathbb{E}[V_T(\boldsymbol{\omega}(a))] + (1-\alpha)\, \mathbb{E}[z] = \bar{a}_{f_j}\, z(a^G, f_i, f_j, e),$$

using $z(a^B, \cdot) = 0$. These objects depend only on primitives, so they are the same in every recursive equilibrium and do not depend on ψ. Hence $\tilde{V}_T(\boldsymbol{\omega}, f_j')$ does not depend on ψ and, by Lemma A.1, neither does market tightness, which is uniquely determined: $\theta_T(\boldsymbol{\omega}, f_j', x, \psi) = \theta_T(\boldsymbol{\omega}, f_j', x)$.

By Lemma A.2, the search values of unemployed and employed workers are

$$\begin{aligned} R_T(\boldsymbol{\omega}) &= \max_{f_j' \in \mathcal{F}(f_i)} \Pi\big(\tilde{V}_T(\boldsymbol{\omega}, f_j') - U_T(\boldsymbol{\omega});\, k_{f_j'}\big), \\ S_T(\boldsymbol{\omega}(a)) &= \max_{f_j' \in \mathcal{F}(f_i)} \Pi\big(\tilde{V}_T(\boldsymbol{\omega}(a), f_j') - V_T(\boldsymbol{\omega}(a));\, k_{f_j'}\big), \qquad a \in \{a^0, a^G, a^B\}. \end{aligned}$$

For each field the optimal tightness is θ^* of Lemma A.2 and, when $\theta^* > 0$, the offered value is $x = \tilde{V}_T - k_{f_j'}/q(\theta^*)$. The field is chosen from at most two elements, so the set of optimal fields exists and does not depend on ψ. Finally, a match that reaches period T is dissolved with probability d that solves

$$\max_{d \in [\delta_{f_j}, 1]} \Big\{ d\, U_T(\boldsymbol{\omega}) + (1-d)\big[V_T(\boldsymbol{\omega}(a)) + \lambda_e S_T(\boldsymbol{\omega}(a))\big] \Big\}.$$

The objective is linear in d: the solution is $d = \delta_{f_j}$ if $V_T + \lambda_e S_T > U_T$, $d = 1$ if $V_T + \lambda_e S_T < U_T$, and any $d \in [\delta_{f_j}, 1]$ under equality ($d = 1$ under the tie-breaking rule). The optimal set does not depend on ψ.

Step 2: induction from $t+1$ to t. Let $H(t)$ denote the statement that U_t, V_t, R_t, S_t, θ_t and the sets of optimal search, separation and acceptance choices at t do not depend on ψ and are the same in every recursive equilibrium. Step 1 establishes $H(T)$. Suppose $H(t+1)$ holds for some $t < T$.

(a) Value of unemployment. From Section 3.2,

$$\begin{aligned} U_t(\boldsymbol{\omega}, \psi) &= b_t + \beta\, \mathbb{E}_{\hat{\psi}|\psi}\big[U_{t+1}(\boldsymbol{\omega}) + \lambda_u R_{t+1}(\boldsymbol{\omega})\big] \\ &= b_t + \beta\big[U_{t+1}(\boldsymbol{\omega}) + \lambda_u R_{t+1}(\boldsymbol{\omega})\big], \end{aligned}$$

since by $H(t+1)$ the terms inside the expectation do not depend on $\hat{\psi}$.

(b) Joint value with known quality. Let

$$W_{t+1}(\hat{\boldsymbol{\omega}}(a)) = \max_{d \in [\delta_{f_j}, 1]} \Big\{ d\, U_{t+1}(\hat{\boldsymbol{\omega}}(a)) + (1-d)\big[V_{t+1}(\hat{\boldsymbol{\omega}}(a)) + \lambda_e S_{t+1}(\hat{\boldsymbol{\omega}}(a))\big] \Big\},$$

which does not depend on $\hat{\psi}$ by $H(t+1)$. For an overeducated worker the continuation value also includes the exogenous offer,

$$\widetilde{W}_{t+1}(\hat{\boldsymbol{\omega}}) = (1-\pi) W_{t+1}(\hat{\boldsymbol{\omega}}) + \pi \max\{W_{t+1}(\hat{\boldsymbol{\omega}}), W_{t+1}(\hat{\boldsymbol{\omega}}^{\pi})\},$$

with $\hat{\boldsymbol{\omega}}^{\pi} = (a, f_i, f_i, \lfloor \tau_{f_i}(e+1) \rfloor)$, and $\widetilde{W}_{t+1} = W_{t+1}$ otherwise; it is the maximum of two objects that do not depend on $\hat{\psi}$, and the acceptance decision is a comparison between them. For $a \in \{a^G, a^B\}$, $V_t(\boldsymbol{\omega}(a), \psi) = z(a, f_i, f_j, e) + \beta\, \widetilde{W}_{t+1}(\hat{\boldsymbol{\omega}}(a))$.

(c) Joint value with unknown quality. Using (b) at the same date t,

$$\begin{aligned} V_t(\boldsymbol{\omega}(a^0), \psi) = \alpha\big[\bar{a}\, V_t(\boldsymbol{\omega}(a^G)) + (1-\bar{a})\, V_t(\boldsymbol{\omega}(a^B))\big] \\ + (1-\alpha)\, \bar{a}\, z(a^G, f_i, f_j, e) + (1-\alpha)\, \beta\, \widetilde{W}_{t+1}(\hat{\boldsymbol{\omega}}(a^0)), \end{aligned}$$

with $\bar{a} = \bar{a}_{f_j}$, which does not depend on ψ.

(d) Market tightness at t. By (c), $\tilde{V}_t(\boldsymbol{\omega}, f_j') = V_t(\nu(\boldsymbol{\omega}, f_j'))$ does not depend on ψ. By Lemma A.1, the zero-profit condition at t determines $\theta_t(\boldsymbol{\omega}, f_j', x)$ uniquely, and it does not depend on ψ.

(e) Search values at t. By Lemma A.2 and (a)–(d),

$$\begin{aligned} R_t(\boldsymbol{\omega}) &= \max_{f_j' \in \mathcal{F}(f_i)} \Pi\big(\tilde{V}_t(\boldsymbol{\omega}, f_j') - U_t(\boldsymbol{\omega});\, k_{f_j'}\big), \\ S_t(\boldsymbol{\omega}(a)) &= \max_{f_j' \in \mathcal{F}(f_i)} \Pi\big(\tilde{V}_t(\boldsymbol{\omega}(a), f_j') - V_t(\boldsymbol{\omega}(a));\, k_{f_j'}\big), \qquad a \in \{a^0, a^G, a^B\}, \end{aligned}$$

together with the corresponding optimal tightness and field choices, do not depend on ψ. In S_t the experience retained at a move is computed from the experience of the searching state, which for a worker employed in $t-1$ is $\hat{\boldsymbol{\omega}}$ and therefore includes her last period of work.

(f) Separation at t. The separation problem at t is the one of Step 1 with U_t, V_t and S_t in place of U_T, V_T and S_T; by (a)–(e) its optimal set does not depend on ψ.

All objects at t are obtained from the primitives and from the objects at $t+1$ through equations with unique solutions, so they are the same in every recursive equilibrium. Hence $H(t)$ holds.

Conclusion. By induction, $H(t)$ holds for every $t \in \{0, 1, \ldots, T\}$, which proves (i) and (ii). For (iii), the construction defines U_t, V_t, R_t, S_t and θ_t for every t; the optimal choices exist because tightness is chosen as in Lemma A.2, the field from a finite set, and the separation probability from a compact interval. Selecting policies with the tie-breaking rule yields policy functions of $(\boldsymbol{\omega}, t)$ alone. Given these policies, the distribution ψ evolves through the law of motion they induce, which is well defined for any initial distribution; since no agent's problem depends on ψ, no fixed point in the space of aggregate states is required. The resulting collection is a recursive equilibrium, it is block recursive, and it is unique under the tie-breaking rule. Without a tie-breaking rule, uniqueness holds for the value functions and market tightness, and for the policy functions at every state in which the worker is not exactly indifferent between two choices. ■

B Construction of the Monthly Panel

This appendix describes how the monthly panel of Section 2.1 is built from the NLSY79. The procedure follows Baley et al. (2022); the code is in the replication package.

Weekly records. The NLSY79 work-history files report, for every week since January 1978, a labor-force status code and, when the respondent works, the numbers of up to five jobs held in that week, together with the usual weekly hours of each job (top-coded at 96). Non-work status codes distinguish unemployment, out of the labor force, active military service, not working with undetermined status, and weeks with no information. From 1994 onwards interviews are biennial, but each interview collects the weekly history since the date of the previous interview, so the years without an interview are covered retrospectively. The panel ends in December 2020. Weekly hours are missing for some jobs, in particular for the job also reported in the Current Population Survey section of the questionnaire in 1980–1993. For these jobs, weekly hours are imputed as the respondent's total weekly hours at all jobs, also reported in the work history, minus the hours of her other jobs in that week, split equally when several jobs have missing hours.

From weeks to months. Each week is assigned to the calendar month in which it starts (its Sunday). For each week, the dominant job is the job with the most hours.

For each month, hours are summed by activity: the hours of each dominant job, and 38 hours for each week of unemployment, non-participation or military service. Weeks with no information, with undetermined non-work status, or in which the respondent is attached to an employer but the periods not worked are not reported contribute zero hours. The activity of the month is the one with the most hours; ties are rare (about 0.5% of months) and are broken by the order of the status codes. A worker is therefore employed in a month when her main job accounts for more hours than any non-work activity, and her job in that month is the job with the most hours. When several jobs are held in the same month, only the job with the most hours is retained.

Job identifiers and tenure. NLSY79 job numbers are assigned within each interview. I link them across interviews using the survey's report of which job at the current interview corresponds to which job at the previous interview, which yields a permanent job identifier. A job not linked to any of the five jobs recorded at the previous interview is treated as a new job. Tenure is the number of months since the start of the current job spell, which starts when the dominant job identifier changes. It is computed on the full monthly panel, before the sample restrictions below, so that it also counts the months of the same job that are dropped for missing occupation.

Occupations. Occupations are reported for each job at each interview, in Census 1970 codes up to the 2000 interview and in Census 2000 codes afterwards. In 1980–1993 the occupation of the job also reported in the Current Population Survey section is recorded only in that section, and it is taken from there. Both are mapped into the harmonized classification of Autor and Dorn (2013), and then into O*NET Job Zones and job families as described in Section 2.2. The seven codes of this classification without any O*NET occupation (farmers, military occupations, unidentified occupations and four small residual codes) are classified as non-college occupations. When the occupation of a job is missing in a month, it is filled with the occupation reported for the same job in the closest later month, that is, at the next interview, or in the closest earlier month if the job is not observed later. Employed months whose occupation is still missing are dropped. In months without a job, the occupation field f_j is that of the most recent job.

Wages. The wage of a month is the hourly rate of pay reported for the dominant job. It is converted into a monthly wage assuming 160 hours per month, deflated by the annual CPI-U (2000 dollars), and expressed relative to the monthly federal minimum wage in 2000 (5.15×160 dollars). Real hourly wages below 1 or above 200 dollars of 2000 are treated as coding errors and set to missing. Wage moments use employed

months with a valid wage (95.3% of employed months).

Employment states. A month is classified as employment (E) when its activity is a job (92.0% of monthly observations). Months of unemployment (2.7%), out of the labor force (4.5%), and not working with undetermined status (0.3%) are pooled into a single non-employment state, labeled unemployment (U) in the paper, because the model has no participation margin. The remaining months (0.6%), mostly months without any reported activity, are classified as neither E nor U, and transitions into or out of them are not counted.

Transitions, gaps and attrition. All transition rates are monthly and are computed between two consecutive calendar months of the same individual. The final sample contains 471 gaps, in 342 of the 2,672 individuals, all created by dropping the employed months without occupation; transitions across a gap are not counted. Apart from the hours and occupation imputations described above, only education is imputed: a missing highest degree is filled in from the adjacent survey rounds and, when it is missing in all of them, imputed from the years of completed schooling, so that respondents with 16 or more years of schooling are classified as college graduates. Individuals who leave the survey contribute observations up to their last month with a reported activity; later months, which no interview covers, are dropped. The UE rate is the share of workers in U who are employed in the following month, the EU rate is the share of employed workers who are in U in the following month, and the cross-field EE rate is the share of employed workers whose dominant job identifier and model job field f_j (Section 4.1) both change between the two months.

Weights. All statistics are weighted by the NLSY79 cross-sectional sampling weight of the survey round held in the calendar year of the month; months in years without an interview, whose history is collected at the next interview, take the weight of the next round in which the respondent is interviewed. Only 0.07% of monthly observations have a zero weight and do not contribute to weighted statistics.

Sample. Starting from the 6,111 respondents of the cross-sectional sample, I drop 272 individuals with more than 24 months of active military service, 1,577 individuals with more than 120 months out of the labor force or without any reported activity since 1979, and 1,352 individuals who worked more than 1,200 hours in 1979, which leaves 2,910 individuals. Of these, 2,695 are observed between ages 30 and 55. I restrict the panel to these ages, drop the 23 college graduates whose major is not reported, and

drop the 2,116 employed months without occupation information. The final sample contains 2,672 individuals and 723,999 monthly observations.

C Calibration Details

Table A1 reports, for each of the 42 targets of Step 2 of the calibration, the data moment, its standard error, the model moment and the t-statistic of their difference.

Table A1: Calibration targets: model and data.

	Data	(s.e.)	Model	t
Calibration targets (42 moments)				
UE rate, non-graduates				
Non-college, ages 30–39	0.1257	(0.0049)	0.1275	+0.4
Non-college, ages 40–49	0.1112	(0.0053)	0.1067	-0.8
Non-college, ages 50–55	0.0750	(0.0047)	0.0763	+0.3
UM rate (unemployed to college job)				
Humanities, ages 30–39	0.1092	(0.0135)	0.1256	+1.2
Humanities, ages 40–49	0.1437	(0.0203)	0.1198	-1.2
Humanities, ages 50–55	0.1198	(0.0236)	0.1047	-0.6
Business/law, ages 30–39	0.1028	(0.0101)	0.1089	+0.6
Business/law, ages 40–49	0.0955	(0.0117)	0.0864	-0.8
Business/law, ages 50–55	0.0745	(0.0132)	0.0731	-0.1
STEM, ages 30–39	0.1097	(0.0153)	0.1089	-0.0
STEM, ages 40–49	0.0783	(0.0112)	0.0864	+0.7
STEM, ages 50–55	0.0886	(0.0166)	0.0696	-1.1
OM rate (overeducated to college job) by field and age				
Humanities, ages 30–39	0.0188	(0.0019)	0.0161	-1.4
Humanities, ages 40–49	0.0079	(0.0013)	0.0104	+1.9
Humanities, ages 50–55	0.0077	(0.0018)	0.0073	-0.2
Business/law, ages 30–39	0.0183	(0.0016)	0.0184	+0.0
Business/law, ages 40–49	0.0074	(0.0008)	0.0080	+0.7
Business/law, ages 50–55	0.0053	(0.0009)	0.0052	-0.1
STEM, ages 30–39	0.0255	(0.0027)	0.0244	-0.4
STEM, ages 40–49	0.0081	(0.0014)	0.0089	+0.6
STEM, ages 50–55	0.0063	(0.0013)	0.0051	-0.9
OU rate (overeducated to unemployment) by age				
All fields, ages 30–39	0.0116	(0.0009)	0.0106	-1.2
All fields, ages 40–49	0.0086	(0.0009)	0.0087	+0.2
All fields, ages 50–55	0.0082	(0.0014)	0.0082	-0.1
EU rate by job tenure				
Non-college jobs (non-graduates), tenure 1–12	0.0246	(0.0008)	0.0244	-0.2
Non-college jobs (non-graduates), tenure 13–36	0.0130	(0.0004)	0.0132	+0.4
Non-college jobs (non-graduates), tenure 37+	0.0057	(0.0002)	0.0055	-1.2

Continued on next page

Table A1 (continued)

	Data	(s.e.)	Model	t
EU rate by job tenure, overeducated graduates				
Non-college jobs, tenure 1–12	0.0261	(0.0024)	0.0242	-0.8
Non-college jobs, tenure 13–36	0.0096	(0.0009)	0.0112	+1.8
Non-college jobs, tenure 37+	0.0043	(0.0004)	0.0051	+2.0
Job separation rate by job tenure				
Humanities jobs, tenure 1–12	0.0521	(0.0031)	0.0394	-4.2
Humanities jobs, tenure 13–36	0.0196	(0.0015)	0.0238	+2.9
Humanities jobs, tenure 37+	0.0101	(0.0007)	0.0089	-1.7
Business/law jobs, tenure 1–12	0.0304	(0.0023)	0.0349	+2.0
Business/law jobs, tenure 13–36	0.0223	(0.0012)	0.0206	-1.4
Business/law jobs, tenure 37+	0.0082	(0.0005)	0.0085	+0.5
STEM jobs, tenure 1–12	0.0267	(0.0025)	0.0334	+2.6
STEM jobs, tenure 13–36	0.0203	(0.0014)	0.0196	-0.5
STEM jobs, tenure 37+	0.0081	(0.0006)	0.0084	+0.6
Log wage change at the move overeducated to college job, relative to stayers				
Humanities, all ages	0.1386	(0.0533)	0.1046	-0.6
Business/law, all ages	0.0420	(0.0380)	0.0528	+0.3
STEM, all ages	0.0593	(0.0403)	0.0497	-0.2

Note: Data: NLSY79, ages 30–55, weighted by the cross-sectional sampling weights; standard errors clustered by worker. Model: calibrated model, computed at the empirical distribution of workers across fields, experience and job tenure at each age. t: (model − data)/s.e. All rates are monthly; job tenure is in months; wage changes are differences in log wages, relative to the wage updates of overeducated graduates who stay in the same job (Table 5). UE: unemployment to employment; UM: unemployment to a matched job; OM: overeducation to a matched job; OU: overeducation to unemployment; EU: employment to unemployment.

Table A2 reports, for each Step 2 parameter, the three targets that contribute most to the precision of its estimate, following Andrews et al. (2017). With the Jacobian G of the model moments and the weighting matrix W of the estimation, the sensitivity of the estimates to the targets is $\Lambda = -(G'WG)^{-1}G'W$. Standardized, $\Lambda_{jk}\hat{\sigma}_k/\mathrm{se}(\theta_j)$ is the change in parameter j, in standard errors, when target k moves by one standard error; its squares sum to one over the targets, and each is the share of the sampling variance of the estimate that comes from target k, ignoring the covariance between targets. The shares confirm the identification argument of Section 4.4: vacancy costs are pinned down by the job-finding rates, learning and exogenous separations by the tenure profiles of separations, the experience kept at a move by the wage changes at exits from overeducation, and the exogenous offers by the OM rates at ages 50–55.

Wages of matches of unknown quality. With complete contracts, the model does not pin down the wage of a match whose quality is still unknown (Section 3). The paper pays every employed worker a constant share of the output of a good match,

Table A2: Sensitivity of the Step 2 estimates to the targets.

Parameter	First target	Second target	Third target
δ_0	EU, non-graduates, 37+ months (63%)	EU, overeducated, all fields, 37+ months (19%)	EU, non-graduates, 13–36 months (5%)
δ_c	Separations, Business/law, 37+ months (38%)	Separations, STEM, 37+ months (27%)	Separations, Humanities, 37+ months (13%)
λ_e	OM, STEM, 30–39 (19%)	OU, all fields, 30–39 (15%)	OM, STEM, 40–49 (15%)
α	EU, non-graduates, 13–36 months (27%)	EU, non-graduates, 1–12 months (22%)	Separations, Business/law, 1–12 months (16%)
k_0	UE, non-graduates, 30–39 (68%)	UE, non-graduates, 40–49 (20%)	UE, non-graduates, 50–55 (7%)
k_1	UM, Humanities, 30–39 (59%)	UM, Humanities, 40–49 (21%)	UM, Humanities, 50–55 (8%)
k_2	UM, Business/law, 30–39 (15%)	UM, Business/law, 40–49 (15%)	Separations, Business/law, 13–36 months (11%)
k_3	Separations, STEM, 13–36 months (28%)	UM, STEM, 40–49 (18%)	Separations, STEM, 1–12 months (16%)
$\bar{a}_0$	EU, non-graduates, 13–36 months (57%)	EU, non-graduates, 37+ months (8%)	EU, non-graduates, 1–12 months (7%)
$\bar{a}_1$	Separations, Humanities, 13–36 months (61%)	Separations, Humanities, 1–12 months (13%)	Separations, Business/law, 37+ months (5%)
$\bar{a}_2$	Separations, Business/law, 13–36 months (18%)	OM, Business/law, 40–49 (12%)	OM, STEM, 50–55 (7%)
$\bar{a}_3$	Separations, STEM, 13–36 months (55%)	Separations, STEM, 1–12 months (26%)	Separations, Business/law, 37+ months (6%)
g_b	UE, non-graduates, 50–55 (49%)	UE, non-graduates, 30–39 (21%)	EU, non-graduates, 13–36 months (17%)
π	OM, Business/law, 50–55 (22%)	OM, STEM, 50–55 (16%)	Wage change at OM, Business/law (15%)
τ_1	Wage change at OM, Humanities (41%)	OM, Humanities, 50–55 (17%)	OM, Humanities, 30–39 (16%)
τ_2	Wage change at OM, Business/law (80%)	OM, STEM, 50–55 (5%)	UM, Business/law, 30–39 (2%)
τ_3	Wage change at OM, STEM (77%)	OU, all fields, 30–39 (3%)	OM, STEM, 50–55 (3%)
χ_1	Separations, Humanities, 13–36 months (32%)	UM, Humanities, 40–49 (11%)	Separations, Humanities, 1–12 months (10%)
χ_2	UM, Business/law, 30–39 (21%)	OM, STEM, 30–39 (13%)	Wage change at OM, Business/law (11%)
χ_3	Separations, STEM, 13–36 months (23%)	OM, STEM, 40–49 (20%)	Separations, STEM, 1–12 months (13%)

Note: For each parameter of Table 8, the three targets with the largest share of the sampling variance of the estimate (in parentheses), from the standardized sensitivity $\Lambda_{jk}\hat{\sigma}_k/\mathrm{se}(\theta_j)$ of Andrews et al. (2017), with $\Lambda = -(G'WG)^{-1}G'W$ and G and W as in Table 8. Age bands for the transition rates; job-tenure bins for the EU and separation rates. Abbreviations as in Table A1.

so that the wage data of Step 1 and the wage changes at exits from overeducation of Step 2 map into the output of a good match. Table A3 compares this convention with the alternative of paying a share of the expected output, $\bar{a}_{f_j}\, y(a^G, f_i, f_j, e)$, while the quality is unknown, with the calibrated parameters. Under the alternative, a graduate who leaves overeducation enters a college match of unknown quality and is paid only the share $\bar{a}_{f_i}$ of its output, so the model predicts wage losses of 38–42 log points at these transitions, and of 11–39 log points even if all experience were kept at the move ($\tau_{f_i} = 1$), against gains of 4–14 log points in the data. The alternative also implies that the wages of surviving matches in college jobs rise by 44–53 log points over the first 24 months of tenure, as their good quality is revealed, while in the NLSY79 the wage of graduates who stay in the same college job grows by 1.9–3.1 log points per year in the first three years of tenure and by 1.0–1.7 log points per year afterwards. Both predictions are rejected by the data, which supports the convention of the paper.

D Additional Decomposition Results

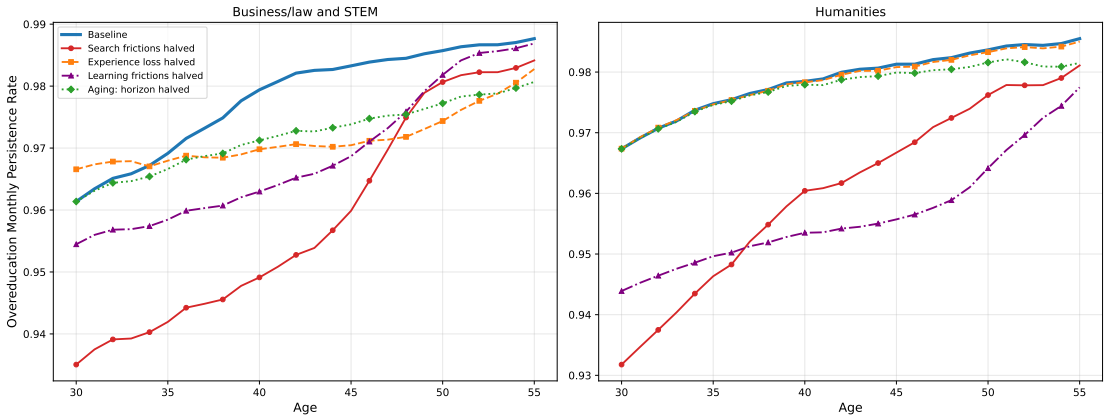
Figure A1 and Figure A2 report the monthly OO rate by age and the survival of the cohort overeducated at 30 when one channel at a time is halved. Figure A3 and Table A4 report the decomposition when the channels are removed entirely. Figure A4 splits the OM rate of the calibrated economy into moves through search and accepted offers.

Table A3: Wages of matches of unknown quality: wage change at exits from overeducation under two conventions.

	Data		Model: wage change at the move			Model: wage growth,
	Wage change at the move	(s.e.)	Good-match wages	Expected-output wages	Expected-output, $\tau_{f_i} = 1$	first 24 months, expected-output wages
Humanities	0.139	(0.053)	0.105	-0.418	-0.393	0.526
Business/law	0.042	(0.038)	0.053	-0.382	-0.149	0.460
STEM	0.059	(0.040)	0.050	-0.381	-0.109	0.436

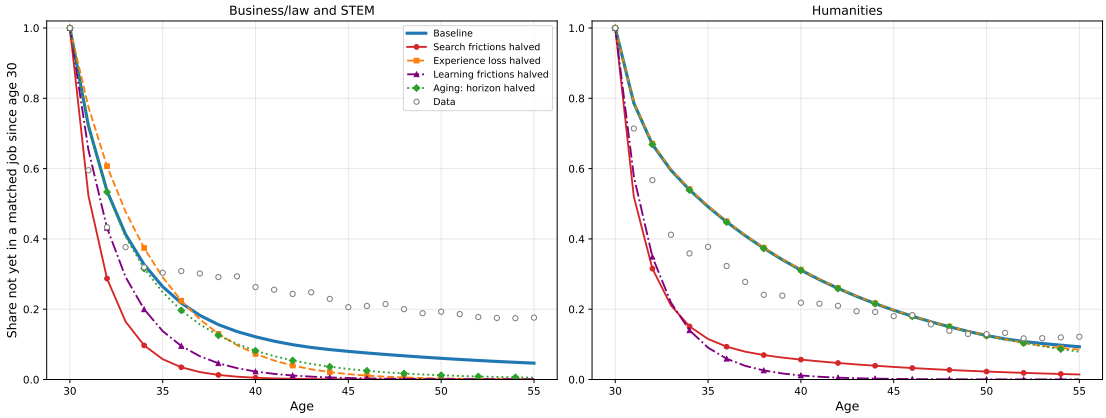
Note: Rows: field of study of the graduates. Data: change in the log wage at transitions from overeducation to a matched job, relative to overeducated stayers (Table 5), with standard errors clustered by worker. Model, with the calibrated parameters of Tables 6 and 8: change in the log wage at these transitions, averaged over the transitions predicted by the model (as for the calibration target), when the wage is a constant share of the output of a good match whether or not the quality is known ("Good-match wages", the convention of the paper), and when it is a constant share of the expected output $\bar{a}_{f_j}\, y(a^G, f_i, f_j, e)$ while the quality is unknown ("Expected-output wages"), with the calibrated share τ_{f_i} of experience kept at the move or with all of it. Last column: increase in the log wage between the first and the 24th month of tenure of the surviving matches in college jobs of the field, implied by expected-output wages (share of survivors whose good quality has been revealed times $-\ln \bar{a}_{f_i}$). All changes are differences in log wages.

Figure A1: Monthly OO rate by age, one channel halved.



Note: Monthly probability that an overeducated graduate is still overeducated in the following month (OO rate), by age, for business/law and STEM graduates pooled (left) and humanities graduates (right). Thick line: calibrated economy. Other lines: economies in which a single channel is halved (Section 6.1) and the others are kept at their calibrated values. All rates are computed at the empirical distribution of experience and job tenure of overeducated graduates at each age.

Figure A2: Cohort of graduates overeducated at age 30, share not yet in a matched job, one channel halved.



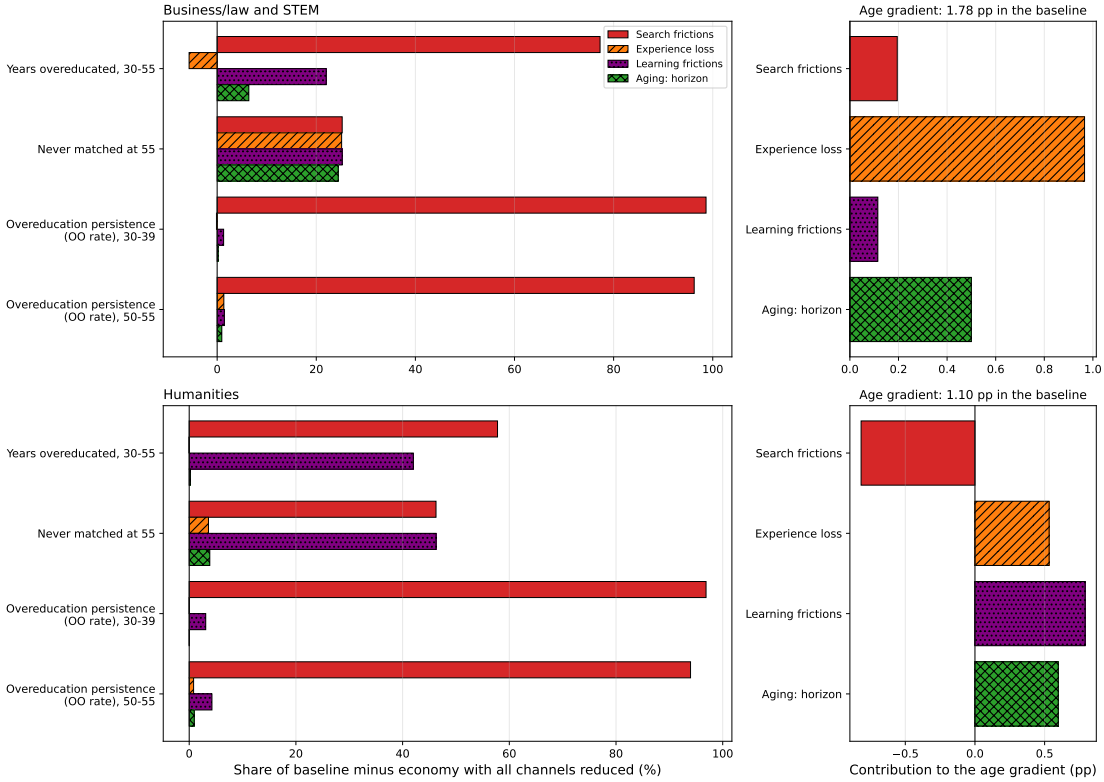
Note: Graduates overeducated at age 30: share that has not held a matched job since age 30, by age, for business/law and STEM graduates pooled (left) and humanities graduates (right). Thick line: calibrated economy, as in Figure 7. Other lines: economies in which a single channel is halved (Section 6.1) and the others are kept at their calibrated values; the cohort starts from the same distribution of experience and job tenure at age 30 in every economy. Hollow markers: NLSY79 data, as in Figure 1.

Table A4: Shapley decomposition: channels halved and removed.

	Humanities				Business/law and STEM			
	Years		Gradient		Years		Gradient	
	Halved	Removed	Halved	Removed	Halved	Removed	Halved	Removed
Search frictions	46%	58%	-0.88 pp	-0.82 pp	59%	77%	-0.51 pp	+0.19 pp
Experience loss	1%	-0%	+0.14 pp	+0.53 pp	-0%	-6%	+1.49 pp	+0.97 pp
Learning frictions	53%	42%	+0.47 pp	+0.79 pp	33%	22%	+0.16 pp	+0.12 pp
Aging: horizon	1%	0%	+0.51 pp	+0.60 pp	8%	6%	+1.10 pp	+0.50 pp
Baseline	7.7	7.7	1.10 pp	1.10 pp	4.4	4.4	1.78 pp	1.78 pp
All channels reduced	1.1	0.1	0.87 pp	-0.00 pp	1.5	0.1	-0.46 pp	-0.00 pp

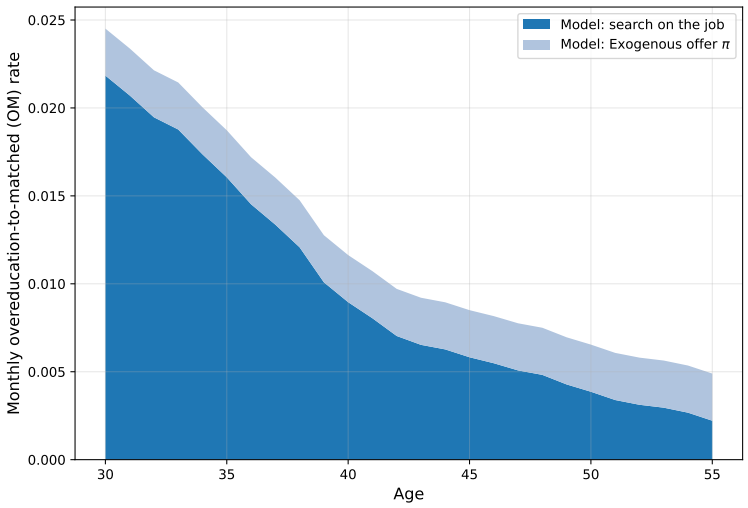
Note: Shapley decomposition of two measures of persistence, for humanities graduates and for business/law and STEM graduates pooled: Years, expected years spent in overeducation between ages 30 and 55 by the cohort of graduates overeducated at age 30; Gradient, monthly OO rate at ages 50–55 minus OO rate at ages 30–39, computed at the empirical distribution of experience and job tenure. Two exercises: "Halved", every channel is halved, as in Table 10; "Removed", every channel is removed entirely ($s = 100\%$). The last two rows report each measure in the calibrated economy ("Baseline") and in the economy with all channels halved or removed ("All channels reduced"); the rows above attribute the difference between the two to each channel, as a share of the difference for Years and in percentage points for the gradient.

Figure A3: Shapley decomposition of persistence, channels removed entirely.



Note: Same as Figure 9, but each channel is removed entirely ($s = 100\%$ in Section 6.1) instead of halved, and the Shapley values are computed over the 16 economies in which each channel is either at its calibrated value or removed. Business/law and STEM graduates pooled (top) and humanities graduates (bottom). Left panels: share of the difference between the calibrated economy and the economy with all channels removed attributed to each channel, for each measure of persistence; the shares of each measure sum to 100%. Right panels: contribution of each channel to the age gradient of the OO rate (OO rate at 50–55 minus OO rate at 30–39), in percentage points. Without search frictions every searching worker finds a job, so the OO rates fall far below their calibrated values and are attributed almost entirely to search frictions.

Figure A4: OM rate by age in the calibrated economy: moves through search and exogenous offers.



Note: Monthly probability that an overeducated graduate moves to a matched job in the following month (OM rate), by age, in the calibrated economy, for all graduates pooled (the three fields weighted by their shares of overeducated employment). The OM rate is split into moves through on-the-job search (dark) and accepted exogenous offers π (light); the two areas are stacked, so the top of the light area is the total OM rate. Rates are computed at the empirical distribution of experience and job tenure at each age.